LPS 105a/205a, PHIL 105a/205a, LSCI 145a: Set Theory and Mathematical Reasoning

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Set theory is the branch of mathematics that makes precise our ordinary reasoning about collections of things. For much of the history of mathematics, there was very little formal study of sets—the relevant ideas were just treated intuitively. But starting in the 1870s, Georg Cantor began explicitly reasoning about the properties of infinite sets. His work radically reshaped philosophical and mathematical understandings of infinity in ways that were quite controversial for several decades.

In the early 20th century, set theory became an important branch of mathematics. While calculus, geometry, and linear algebra are the branches of mathematics that have had the most applicability in much of science and engineering, set theory (and related areas of logic) has been the most important branch of mathematics in philosophy and linguistics.

In mathematics, set theory has primarily become relevant as a framework within which all of the rest of mathematics is done. Sets are essential for the concept of number—what you are counting when you count is a *set* of things. Gottlob Frege's *Grundlagen der Arithmetik* (1884) (The Foundations of Arithmetic) and *Grundgesetze der Arithmetic* (vol. 1, 1893; vol. 2, 1903) (The Basic Laws of Arithmetic) showed how to define the counting numbers as certain sets of sets, and Richard Dedekind's *Was Sind und Was Sollen die Zahlen?* (1888) (What are the numbers, and what should they be?) showed how to define other types of numbers, like rational and real numbers, as sets of counting numbers. Although Bertrand Russell found a contradiction in Frege's set theory, Russell and Alfred North Whitehead developed a new set theory in their *Principia Mathematica* (vol. 1, 1910; vol. 2, 1912; vol. 3, 1913) and showed how to formulate most of the currently existing mathematics inside of it. Bartel Leendert van der Waerden's *Moderne Algebra* (vol. 1, 1930; vol. 2, 1931) reformulated all of abstract algebra in a set-theoretic language, and his use of the Axiom of Choice led to many new developments in mathematics, but also various controversies. But in recent decades, most mathematicians have found it useful to jump directly from some simple, intuitive set theory to the Axiom of Choice, and set theory is no longer considered as central and important as number theory, algebra, topology, geometry, and analysis.

In linguistics, set theory has been central to the development of semantics (the study of systematic linguistic meaning). It has also been relevant to pragmatics (the study of the purposes for which speakers use linguistic meaning) and syntax (the study of the systematic arrangement of words), but has been less important in morphology, phonology, and phonetics. This work began with Frege's *Begriffsschrift* (1879) (Concept Writing), in which he introduces modern quantifier logic, and uses sets as the meanings of concepts. Alfred Tarski then showed, in "Der Wahrheitsbegriff in den formalisierten Sprachen" (1935, Polish original in 1933) (The concept of truth in formalized languages) how to use a set-theoretic formulation to precisely define what it is for a sentence to be true. While Tarski's work concerned formal languages, Richard Montague showed how to use the same set-theoretic method to give a formal semantics for natural language in a series of three papers, "Universal Grammar" (1970), "English as a Formal Language" (1970), and "The Proper Treatment of Quantification in Ordinary English" (1973). Barbara Partee showed how to generalize this work, and set-theoretic concepts soon became relevant to many other aspects of meaning. Saul Kripke gave a basic treatment of modal verbs ("can", "may", "must", "should", etc.) in terms of sets of possible worlds, which Angelika Kratzer has turned into a fuller theory of conversational meaning, with sets representing not just the meanings of concepts and sets of possible worlds, but also information states, events, and other

important aspects of meaning. Irene Heim used set theory to represent conversational contexts, and showed how meanings can be understood in terms of changes in context rather than sets of worlds.

In philosophy, the importance of set theory primarily comes through its influence in mathematics and linguistics. The work of Frege, Russell, and Tarski resolved many ancient paradoxes around truth and infinity, but also introduced new paradoxes that have not fully been resolved. The role of set theory in providing a foundation for mathematics helped address some questions in the philosophy of mathematics, particularly about the existence of abstract objects, but also raised new ones. The role of set theory in semantics has also become important for philosophy of language, but also philosophy of mind and epistemology. The use of set theory in the understanding of infinity, and modal concepts like necessity and possibility, has taken on great importance in metaphysics, and thinking about the relationship of sets to their members has been relevant for the study of parts and wholes.

Each section of this document corresponds roughly to my plans for one week of the term. Each section will include some definitions and expository writing, some example questions with their answers worked out, and then starting with Week 1, some exercises. I will aim to have student volunteers to present some or all of the exercises in class, and students will be asked to turn in a written answer to one of these exercises each week. Feel free to choose a relatively easy question if you want practice being precise, or a relatively hard one if you want practice managing the steps in a more complex proof.

Part I does set theory informally, in the style of Cantor, leading up to the controversies around the Axiom of Choice, and the paradoxes that doomed Frege's approach to set theory. Although contradictions are possible within the informal set theory that we start with, Cantor was able to skillfully avoid them. In this first part of the class, we will sometimes point out these contradictions, but will carefully back away until it comes time to confront them.

Part II begins after we confront the paradoxes of informal set theory. This traces the development of modern axiomatic set theory, starting with the work of Ernst Zermelo in his 1908 paper, "Untersuchungen über die Grundlagen der Mengenlehre" (Investigations on the foundations of set theory) and its modification by Abraham Fraenkel. In the beginning of this section, we will mostly be re-proving the same results that we proved informally, but this time working within a formalized axiom system. Both formalized and informal proofs are parts of contemporary mathematical practice, so it will be helpful to spend some time with each. We will be able to see how the modern formalized system blocks the specific paradoxes that arose for the informal system (though some of the important results we will be getting close to at the end show that there can be no formal proof that contradictions are impossible, except by using even more questionable formal systems).

New terms that are being introduced will be given in *italics*.

Part I: Informal Set Theory

0 Sets as the meanings of words

0.1 Introduction

Sets are collections of things. It's hard to be more precise about this until we get into axiomatic set theory, but we can work with examples. I'll do this by introducing a bit of terminology. The things that are in a set are called its *members* or *elements*. Everything that matters about a set is determined by which things are members of the set and which things are not members of the set. There are three ways that sets are often presented: entirely in words by saying "The set of ...", or explicitly by listing all the elements between two curly braces, or with the *set-builder notation* by using curly braces, a variable, a vertical line, and a description involving that variable that definitively specifies for each object whether it is a member of the set or not. For instance:

$$\begin{aligned}\text{The set of days of the week} &= \{\text{Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday}\}\\ &= \{x \mid x \text{ is a day of the week}\}.\end{aligned}$$

Although there are many other things that have members, like teams and committees and departments, these are importantly different from sets, in that they can be different even if they have the same members. Maybe there are two committees that both have Sue, John, and Tania as their only members—but if one of the committees has Sue as the chair and is in charge of determining the curriculum, while the other committee has John as the chair and is in charge of hiring, then they are different committees. But they have the same set of members. More clearly:

$$\text{The set of days of the week whose name begins with "S"} = \text{The set of weekend days.}$$

Note also that

$$\{\text{Saturday, Sunday}\} = \{\text{Sunday, Saturday}\} = \{\text{Saturday, Sunday, Saturday}\}.$$

Everything that is a member of one of them is a member of the others, so they are the same set, even though they are written differently.

0.2 Notation

The symbol "$\in$" is used to state that something is a member of a set (or with a slash to say it is not). For instance

$$\begin{aligned}\text{Ireland} &\in \text{The set of members of the European Union}\\ \text{Kenny} &\in \text{The set of faculty in the Department of Logic and Philosophy of Science}\\ \text{Canada} &\notin \text{The set of countries that have Spanish as an official language}\\ 5 &\notin \{x \mid x \text{ is even}\}\end{aligned}$$

There is one very significant set known as the *empty set*. This is the set for which nothing is a member. Although we could notate this set "{}", it's more standard to notate it as "$\emptyset$". Of course, this set also has other names, like "The set of faculty in LPS who are over 100 years old" and "The set of members of the European Union whose capital is in South America". (Why is there only one empty set?)

There are some important relations and operations among sets. If every member of A is also a member of B, we say that A is a *subset* of B or B is a *superset* of A, written as "$A \subseteq B$" or "$B \supseteq A$". Which of the following sets are subsets of each other?

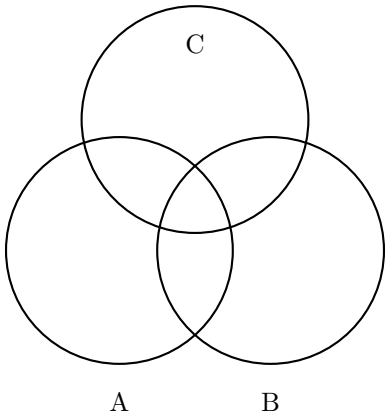
- A =The set of even numbers

- $B = \{2, 6\}$
- $C = \{x \mid x < 7\}$
- $D =$The set of odd numbers
- $E =$The set of all positive integers
- $F = \emptyset$
- $G =$The set of all numbers that are not multiples of 4

Note that every set is a subset and a superset of itself. But if A is a subset of B and is not equal to B, we say A is a *proper subset* of B or B is a *proper superset* of A, and we usually write "$A \subset B$" or "$B \supset A$" with this meaning (like the difference between "$<$" and "$\leq$" or "$>$" and "$\geq$"—though there are some texts that use "$\subset$" to mean "$\subseteq$", which can be confusing).

For any two sets, x and y, we define the *intersection* of x and y ("$x \cap y$") to be the set of all things that are in both sets, and the *union* of x and y ("$x \cup y$") to be the set of all things that are in either of the sets. (You can remember which symbol is which because the $\cup$ is for Union and the $\cap$ is for iNtersection.) What is $A \cap C \cap E \cap G$? What is $B \cup C$? What is $A \cup D$? What is $B \cup F$? What is $B \cap F$?

We say that two sets are *disjoint* if their intersection is empty. Which pairs of sets from the ones above are disjoint?



If we think of each letter as standing for the set of points inside the relevant circle, then which regions are $A \cup B$? $A \cup C$? $B \cup C$? $A \cap B$? $A \cap C$? $B \cap C$? $A \cap (B \cup C)$? $(A \cap B) \cup C$? $(A \cap B) \cup (A \cap C)$? $(A \cup C) \cap (B \cup C)$? $A \cap (B \cap C)$? $(A \cap B) \cap C$? $A \cup (B \cup C)$? $(A \cup B) \cup C$? (If you are familiar with the terms from algebra, you can think about commutativity, associativity, and distributivity.)

0.3 Vagueness and ambiguity

The important thing for a set is that for every object, it is either definitely a member of the set, or definitely not a member of the set. It doesn't matter if anyone knows what the members are, or even could know what the members are. But there must be a definite fact. Consider the set of students in this class who will some day win a Nobel prize or the set of cities that have un-discovered tyrannosaur fossils buried under them or the set of numbers that have never been written down in Arabic notation. These are perfectly fine sets. But some problematic definitions that don't work include definitions like "The set of tall people" or "The set of mountains". (Why are these problematic?)

Paradox 0.1 (The Sorites Paradox) Both of these premises seem plausible:

- A million grains of sand form a heap (σωρειτης).

- A heap of sand minus one grain is still a heap of sand.

But they together lead to the conclusion that there can be a heap of sand with no sand grains at all! (Why is that?)

A word that has some version of the sorites paradox is usually said to be *vague*. This is importantly different from an *ambiguous* word like "star", which has multiple distinct meanings (it can either refer to famous movie actors or to balls of hydrogen undergoing fusion in outer space—note that at least one of these meanings is in fact vague). It is also importantly different from a *polysemous* word like "month", which has several related meanings (it can refer to one of the 12 subdivisions of the year, or to any period of about 30 days). It is common to replace a vague word with a *precisification*, some set that does have precise membership. For instance, we can replace the set of tall people with the set of people who are at least six feet tall, or the set of mountains with the set of geographic peaks that are over 5,000 feet high and not connected to any higher peak without going below that elevation.

Once we have precisified vague terms, we can think of one aspect of the meaning of most adjectives, nouns, and intransitive verbs as being the set of objects that they describe. We call this set the *extension* of the word. That is, the extension of "quick" is the set of quick things, the extension of "brown" is the set of brown things, the extension of "dog" is the set of dogs, the extension of "fox" is the set of foxes, the extension of "jumped" is the set of things that jumped, the extension of "sleeps" is the set of things that sleep.

The extensions of phrases composed out of adjectives and nouns can usually be thought of as the intersection of the extensions of the individual words. So the extension of "lazy fox" is the intersection of the set of lazy things and the set of foxes. The extension of "quick brown dog" is the intersection of the set of quick things, the set of brown things, and the set of dogs. (The extension of the word "the" is somewhat complicated, so we can't get to the extension of the "the quick brown dog" yet!)

However, there are some adjectives that don't work this way. One set of counterexamples includes words like "former", "fake", and "imitation", and some similar words like "suspected" and "alleged". (How would you characterize the similarities and differences between these words?) Another set of counterexamples include words like "typical", "excellent", "bad", and "tall". (What's the difference between Alex, who is a tall six-year-old, a typical boy, an excellent student, and a bad basketball player, and Charlie, who is a tall basketball player, an excellent boy, and a bad student? Could he be a typical six-year-old?)

Paradox 0.2 (Grelling's paradox) Note that some words are a member of their own extension: "word", "pentasyllabic", "English", "familiar". Note that other words are not a member of their own extension: "dog", "monosyllabic", "French", "microscopic". Kurt Grelling invented the words "autologous" for words that are members of their own extension, and "heterologous" for words that are not members of their own extension. Is the word "heterologous" heterologous? (Also, is the word "autologous" autologous?)

1 Sets and numbers

1.1 Natural numbers

What are numbers?

The natural numbers are the things we use to say how many of something there are when there are finitely many of them. $\mathbb{N} = \{0, 1, 2, 3, \dots\}$. Note that we have to include 0, because that is sometimes how many of something there are—it is how many unicorns are in the room as I type this.

What does it mean to say how many of something there are? Well, when do we care how many of something there are?

Think about the set of students in this class and the set of chairs in the room. Do we have the same number of students as chairs, or does one of these sets have more? You don't need to count—just look at whether there are any chairs left over, or any students left over, when everyone is sitting in their own chair!

This is the basic idea of numbers. Two sets have the same number of things when there is a *one-to-one correspondence* between them. We will define it precisely below.

1.2 Correspondences and ordered pairs

What is a one-to-one correspondence between two sets? The best way to think of it is as a special type of set of ordered pairs.

What's an ordered pair? We write it with ordinary parentheses, and its two members. The important thing is that even though $\{x, y\} = \{y, x\}$, $(x, y) \neq (y, x)$, unless $x = y$. You might be familiar with ordered pairs as the coordinates of points in the plane. Remember that the point $(1, 0)$ is 1 unit to the right of the origin, while $(0, 1)$ is 1 unit above the origin. Note that $(0, 0)$, $(1, 1)$, and so on are ordered pairs as well—because of the ordering, it makes a difference if we have the same element twice.

If X and Y are sets, we can consider some set S of ordered pairs whose first member is in X and whose second member is in Y.

Here are the four conditions for a one-to-one correspondence:

- For every $x \in X$, there is at most one $y \in Y$ with $(x, y) \in S$ (Function)
- For every $x \in X$, there is at least one $y \in Y$ with $(x, y) \in S$ (Total)
- For every $y \in Y$, there is at most one $x \in X$ with $(x, y) \in S$ (One-to-one)
- For every $y \in Y$, there is at least one $x \in X$ with $(x, y) \in S$ (Onto)

Why are these the conditions we want for the set of students and chairs? Are there any conditions we don't really care about?

Exercise 1.1 For every set, there is at least one correspondence between it and itself.

Exercise 1.2 For a set with two elements there are precisely two correspondences between it and itself. For a set with three elements there are precisely six correspondences between it and itself.

Exercise 1.3 If there is a correspondence between A and B, and a correspondence between B and C, then there is a correspondence between A and C.

Exercise 1.4 If there is a correspondence between A and B, then there is a correspondence between B and A.

Note that although the existence of a correspondence proves that two sets have the same size, the fact that one attempted correspondence fails doesn't mean that they don't have the same size. Imagine someone came into the room early and claimed 15 chairs for themself, and let everyone else fight over the rest. There would be lots of pairs with the one person attached to different chairs, so the Function condition would fail. For other people, they might either not get a chair (so that the Total condition fails) or might share a chair (so the One-to-one condition fails). But that doesn't mean that there can't be a correspondence—just that this isn't it.

1.3 Correspondences and counting

We don't always try to establish a correspondence to show that two sets have the same number of things. For finite sets, we rarely do. Instead, we count.

What do we do when we count? We say each number in sequence, while picking out one element of the set. The last number we say is the number of things in the set.

We have used the numbers in two different ways—as we are counting, we are picking out a first, second, third, etc. member of the set. These are called *ordinal* number, and they represent positions in an ordering. We use the last number of the counting to say how many members are in the set. This is called a *cardinal* number.

Exercise 1.5 Counting a set establishes a correspondence between it and the natural numbers below its cardinality.

Exercise 1.6 If you can count two sets, and get the same cardinal number, then there is a correspondence between the two sets.

Note that counting only works to establish correspondences for finite sets, but correspondences can exist even for infinite sets. But with infinite sets, some strange things can happen.

Galileo observed that there is a correspondence between the natural numbers and the perfect squares, even though the perfect squares are a proper subset of the natural numbers. Galileo used this to suggest that there is no meaning of "same size" for infinite sets, but Cantor was a bit more careful. He treated this correspondence as meaning that these sets have the same size even though one is a proper subset of the other.

Note that this means that establishing a one-to-one correspondence between X and a subset of Y doesn't necessarily mean that Y is larger than X—it might still be the same size. This failed attempt at a correspondence would just be like the attempt above where one person hogged some of the chairs.

In fact, Dedekind defined an infinite set as one that has a one-to-one correspondence with a proper subset. (Modern set theorists don't exactly use this definition—it turns out that it gets caught up in the controversy around the Axiom of Choice, which we will discuss later.)

1.4 First proof by induction

The natural numbers are the things we use for counting. It doesn't matter if we use "0", "1", "2", "3", ..., or "zero", "one", "two", "three", ..., or "zero", "uno", "dos", "tres", ..., or "null", "eins", "zwei", "drei", ..., or even "eenie", "meenie", "meinie", "moe", ..., as long as there is a specific one to start with, and a distinctive next one at each step. This gives rise to a distinctive procedure we can often use to prove that something is true for every natural number—as long as we show that this is true for the natural number at the start of the list, and whenever it is true for one natural number it is true for the next natural number, then we know it must be true for all natural numbers.

We will use this proof method quite often. Here is the first example.

Theorem 1.1 If a set's cardinality is a natural number, then it does not have a correspondence with any proper subset.

Proof. By induction.

Base case: Prove that a set with 0 elements does not have a correspondence with any proper subset. (Trivial, because there are no proper subsets!) Induction step: Prove that if a set with n elements has no correspondence with any proper subset, then a set with $n+1$ elements does not either. This step takes more work. We show that *if* there *were* a correspondence of a set with $n+1$ elements with some proper subset, *then* we could use that to build a correspondence of a set with n elements to a proper subset. Since that latter correspondence is assumed not to exist, neither can the first. We do this in two cases. Let x be the extra element in our set of $n+1$ elements.

First case: Assume the correspondence between a set with $n+1$ elements and a proper subset left out x in its range. Then that means that x in the domain must have been sent to some element a of the original set in the range. Since it's a correspondence, no other element of the domain can have been sent to a. So delete x from the domain, and now we have a correspondence of the original set to some proper subset of itself, leaving out a.

Second case: Assume the correspondence between a set with $n+1$ elements and a proper subset included x in its range. Then there must be some element a of the original set that is left out of the range. Just deleting x from the domain and range doesn't give a correspondence—some element b of the domain must have been sent to x in the range, and x in the domain must have been sent to some element c of the range. But if we delete x from the domain *and* range, and send b to c, then now we have a correspondence between the original set and a proper subset leaving out a. □

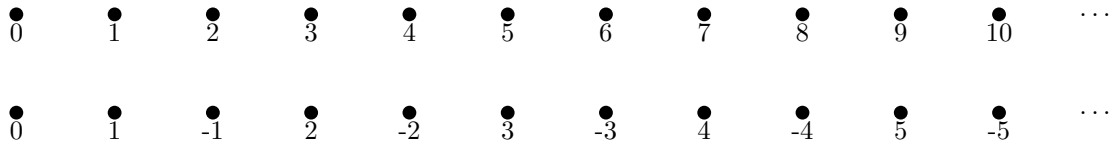
The official definition of a "finite set" is that it is one whose cardinality is a natural number. As we'll see later, we haven't yet precisely defined what the natural numbers are—at best, we've said that they're 0, 1, 2, 3, ..., and everything else you reach by counting. An infinite set is one that doesn't have a natural number as its cardinality.

2 Well-orderings, countable vs uncountable infinities

2.1 Infinite sets, and subsets

Although we haven't shown that every infinite set has a correspondence to a proper subset, some of the early important results by Georg Cantor involved showing that the infinite sets we can think of all do.

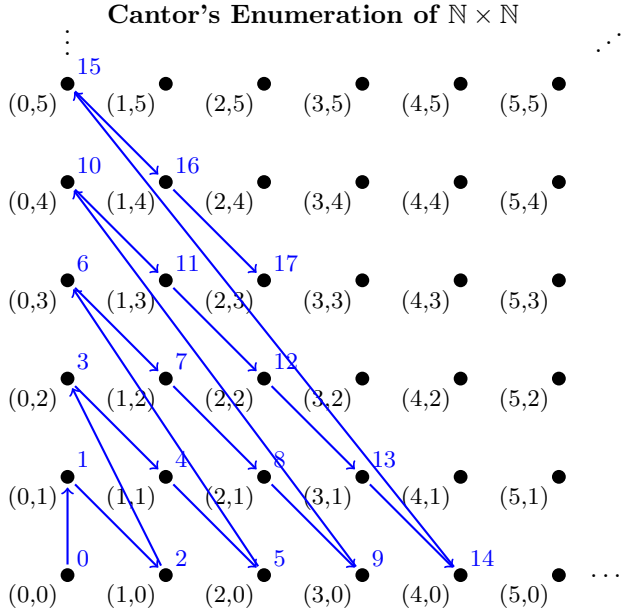
For instance, let $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ be the set of all integers. Although the natural numbers are a proper subset of the integers (i.e., $\mathbb{N} \subset \mathbb{Z}$), there is a correspondence between them. We can line up 0 with 0, and the positive integers with the even natural numbers, and the negative integers with the odd natural numbers, giving a correspondence like this:



It's not hard to see that every natural number, and every integer, occurs exactly once in its relevant row, so this really is a correspondence.

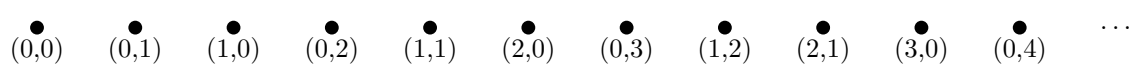
And in fact, we can do this even for sets that seem infinitely many *times* as large as the natural numbers. For instance, consider the set of all ordered pairs of natural numbers, $\mathbb{N} \times \mathbb{N} = \{(m, n) \mid m, n \in \mathbb{N}\}$. (We'll explain the "$\times$" notation later.) Cantor showed that even this set has a correspondence with the natural numbers:

Cantor's Enumeration of $\mathbb{N} \times \mathbb{N}$



Each pair (n, m) gets a unique natural number following the blue path

We can think of this as putting the set in the same ordering as the natural numbers:



These sets of even numbers, natural numbers, integers, and ordered pairs of naturals, all have correspon-

dences with each other, so we say they have the same cardinality, though this cardinality is not any natural number. Putting any of these sets in correspondence with the natural numbers gives a way to "count up" to each of the elements in the set, so we say this cardinality is that of the "countable" infinities, even though we never finish the process of counting of any of them.

2.2 Motivation for well-orderings

For natural numbers, we have already identified two different roles they play in counting. One is the *ordinal* number role, where they represent what *position* in the count an object has. When you count out all the elements of a set, you are effectively giving each element of that set an ordinal number as the first, second, third, etc. member of the set that you are identifying.

The other is the *cardinal* number role, which tells you how many objects there are in the set *as a whole*. This role is only played by the *last* number that you say when you count a set. (Of course, for the infinite sets, as we have considered them so far, there is no last number, but we'll find another way to name the cardinal number.)

If you count up to 5 while counting the chairs in the room, and you count up to 5 when counting the people, then you know that the chairs and the people can be put into a correspondence. That's what it means to say the two sets have the same cardinal number. But the particular correspondence that we know must exist can be derived by using the ordinal numbers—the first person is assigned to the first chair; the second person is assigned to the second chair; and so on.

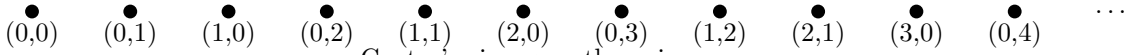
We have seen that for sets whose cardinality is a particular natural number, there is no way to put that set into a correspondence with a proper subset. If a set has 1,352 elements, it is in correspondence with the set $\{1, 2, 3, \ldots, 1350, 1351, 1352\}$. But it can't also be put in correspondence with any proper subset of that set. That means that no matter how you start counting out elements of the set, there will always be some elements left if you haven't gone through all these numbers. And conversely, there will always be numbers from that set left if you haven't gone through all elements of the original set. Any way of giving out ordinal numbers up to the cardinal number will work out just fine.

However, this isn't true when it comes to infinite sets. We have seen that the natural numbers can be put into correspondence with several proper subsets (like the even numbers and the squares) and also with some proper supersets (like the integers) and sets that feel even larger than supersets (like the set of ordered pairs of natural numbers). If we're not careful when giving out ordinal numbers, we could accidentally give them all out before finishing the set we are trying to number. If we count the ordered pairs of natural numbers by just beginning with all the pairs whose second element is 0, we will run out of natural numbers before even getting to the second row of the chart! So we had to do our counting in a different order. Cantor's suggestion involved zigzagging through the chart, but David Hilbert suggested another useful idea for dealing with countable infinities.

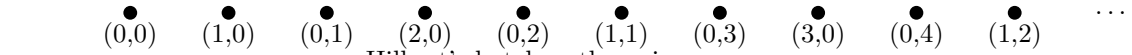
Hilbert noted that if you had an infinite hotel, with one room for every natural number, and two infinite tour buses, each with one guest for every natural number, then if you're trying to give everyone a room, you have to be careful. If you just give everyone from the first bus the room that corresponds to their number, then there's no rooms left for anyone from the second bus. But if you just give out *every other room* when a bus arrives, then you'll always have infinitely many left, and can accommodate more buses if they show up later. So the first bus gets the odd numbered rooms, the second bus gets every other remaining room, the ones that are twice an odd number, and there's still all the multiples of 4 left over for additional buses. This gives another way of giving a correspondence between the natural numbers and the ordered pairs of natural numbers (well, almost—the number 0 never ends up assigned to anything!):



The natural numbers



Cantor's zigzag on the pairs

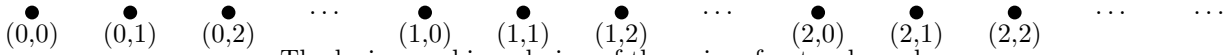


Hilbert's hotel on the pairs

But there are other ways of ordering these infinite sets too. For instance, there is the standard ordering of the integers, and the "lexicographic" ordering of the ordered pairs of natural numbers. (We call it "lexicographic", because this is the principle that a dictionary is organized on—everything with the earliest first letter comes first, then everything with the second-earliest first letter, etc.)



The standard ordering of the integers



The lexicographic ordering of the pairs of natural numbers

Even though both of these orderings are different from the standard ordering of the natural numbers, Cantor noticed an important difference between the two orderings. The ordering on the integers is in a sense "slippery"—if you slide everything over, it still looks the same, so there's no unambiguous way to name the "positions" in the ordering:





But the lexicographic ordering on the pairs of natural numbers is not—even if you delete the names in the positions, you can figure out which position is which. There is still a first, and a second, and a third, and so on. But even after you run out of natural numbers, there's a first position beyond all those (where the pair $(1, 0)$ is going to go) and then a second, and then a third, and so on.

Just as Cantor generalized the concept of a cardinal number to infinite sets by seeing when they have correspondences, he generalized the concept of an ordinal number to positions in infinite orderings that aren't "slippery" in this way. To make this precise, Cantor defined the concept of a *well-ordering*: an ordering called a well-ordering when every non-empty subset has a first element.

2.3 More formal definitions

Intuitively, we sometimes think that we can't "identify positions in an ordering" if you can't count up to there. But Cantor's idea was that to define an ordering, all you need is a set of elements, and a rule that determines, for any two elements, which comes first. (Just like with membership in a set, this rule doesn't need to let us *know* which of two elements comes first—all that matters is that there *is a fact* for any two elements about which one comes first.) For this rule to be an ordering, it needs to be transitive—if x is

before y and y is before z, then x is before z. And it should be irreflexive—nothing should come before itself.

Exercise 2.1 Show that any relation that is transitive (if x is before y and y is before z, then x is before z) and irreflexive (there is no x with x coming before x) must be antisymmetric (if x is before y then y is *not* before x).

We will use the symbol "$<$" for any ordering. Let's be precise and define the *lexicographic* ordering on pairs of natural numbers by saying $(x, y) < (z, w)$ iff $x < z$ or $x = z$ and $y < w$.

Consider the following sets of ordered pairs. What is the earliest element of each of these sets, in the lexicographic ordering?

- $\{(1, 3), (5, 2), (2, 7), (8, 0), (0, 4), (1, 5)\}$
- $\{2, 4), (2, 6), (2, 8), (2, 10), (1, 11), (1, 13), (3, 15)\}$
- The set of all pairs (m, n) where $m = 2n$
- $\{(3, 5), (3, 4), (3, 2), (2, 7), (2, 5), (2, 1), (5, 9), (4, 3), (5, 10)\}$
- $\{(10, 0), (9, 1), (8, 2), (7, 3), (6, 4)\}$

Think about how you found the earliest element of each of these sets.

Exercise 2.2 Show that the lexicographic ordering on ordered pairs of natural numbers is a well-ordering.

Now we can show that no well-ordering is "slippery" in the sense we've talked about.

Theorem 2.1 If a set is well-ordered, then it has no correspondence with itself that preserves the ordering while moving any element.

Proof. By contradiction.

Assume there *was* some correspondence with itself that preserves the ordering, while moving some elements. That is, the set of elements moved by the correspondence is non-empty.

Since this is a non-empty subset of our well-ordered set, there must be an *earliest* element in the ordering that is moved by the correspondence. So, consider whether it could have been moved earlier in the ordering or later.

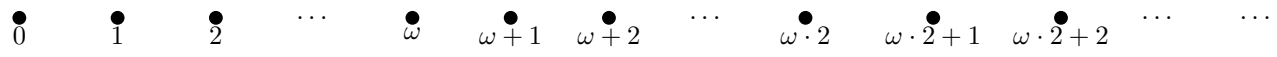
It can't have been moved earlier in the ordering, because this is the earliest element that was moved, so all earlier elements must be sent to themselves. If this had been moved earlier, it would have led to a collision, and the "correspondence" wouldn't be one-to-one.

But it also can't have been moved later in the order. Because this is the earliest element that was moved, anything else that was moved comes later in the ordering, and the correspondence preserves the order. So if it was moved later in the ordering, then all other elements that were moved must have been moved even later. So nothing could have been moved onto this element. So there must be a gap, and the "correspondence" wouldn't be onto.

There is a contradiction either way, so if a set is well-ordered, then no correspondence of the set to itself can preserve the ordering while moving any elements. □

2.4 Ordinals, and their addition

Cantor came up with names for each of the positions in the lexicographic ordering of the pairs of natural numbers, and called them *ordinal numbers*.



Some ordinal numbers

Let's try to explain what these names mean. The first step is just that he uses the letter "ω" to stand for the first infinite position in the well-ordering. But to explain the rest, we need to explain what the symbols "$+$" and "$\cdot$" mean—we are familiar with them for the natural numbers, but we need to explain what they mean in order to see how they generalize to the infinite ordinals.

The way we understand what "$m + n$" means, when m and n are both finite ordinals is by counting m spots, and then counting n more, and seeing what position you get to. (Remember that the ordinal or cardinal of a finite set is the first ordinal you *don't* say when counting up from 0.)

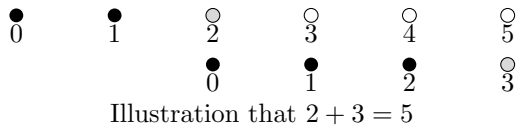
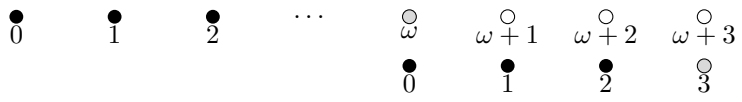


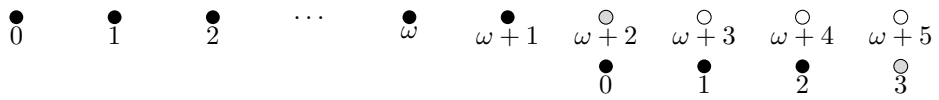
Illustration that $2 + 3 = 5$

And we can do this when one of the terms is infinite. (ω is the ordinal of a set like the natural numbers, where ω is the first ordinal you *don't* say while counting.) Here is what $\omega + 3$ means this way:



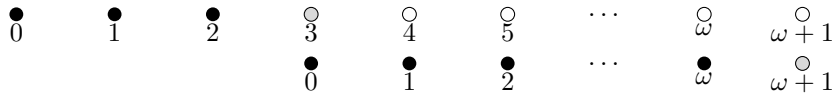
$$\omega + 3 = \omega + 3$$

And adding an infinite ordinal followed by a finite ordinal works as we expect:



$$(\omega + 2) + 3 = \omega + 5$$

But when we do it in the opposite order, some surprising things happen.



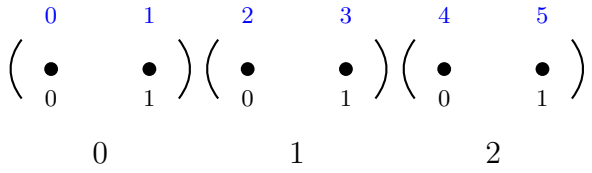
$$3 + (\omega + 1) = \omega + 1$$

Effectively, adding something infinite to the end of something finite "absorbs" the finite thing. If you add something infinite at the end of something infinite with a finite remainder, that finite remainder gets absorbed too.

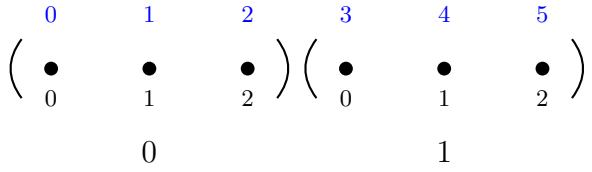
Exercise 2.3 What is $(\omega + 1) + (\omega + 3)$? What is $(\omega + 2) + (\omega + 2)$? What is $(\omega + 3) + (\omega + 3)$? What is $(\omega + 6) + \omega$? Which, if any, of these are equal?

2.5 Multiplication of ordinals

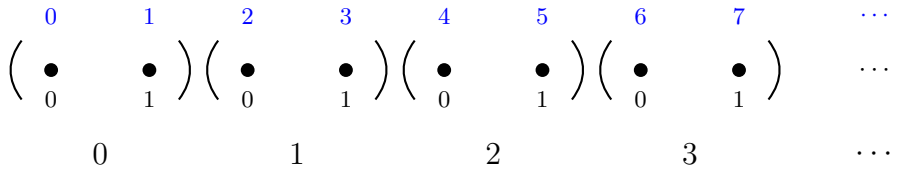
The way we understand what "$m \cdot n$" means, when m and n are both finite ordinals, is by counting m spots, n many times. For instance, this diagram shows that $2 \cdot 3 = 6$ (remembering as always that the name of an ordering is the first ordinal you *don't* say while counting up from 0):



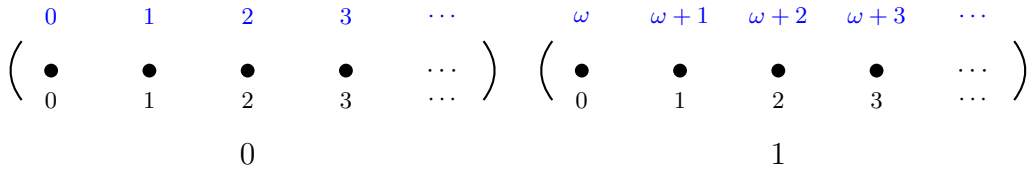
Similarly, this diagram shows that $3 \cdot 2 = 6$:



The same idea works for infinite sets, but with some interestingly different results. Counting 2 spots, ω many times, still uses up only the natural numbers:



But $\omega \cdot 2 = \omega + \omega$. Counting ω spots, 2 times, uses up all the natural numbers before we even start the second group:



Exercise 2.4 What is $2 \cdot (\omega + 3)$? What is $(\omega + 3) \cdot 2$? Is either of them equal to $\omega + \omega + 3 + 3$?

This also means we can multiply larger ordinals, to get things like $\omega \cdot \omega$ (which is the ordering of $\mathbb{N} \cdot \mathbb{N}$ under lexicographic ordering), and $\omega \cdot \omega \cdot \omega$, and so on. You can tell these ordinals are going to get large and complicated.

2.6 Initial segments

An *initial segment* of a well-ordering is a subset such that any element of the well-ordering that is before something in the initial segment is also in the initial segment.

Theorem 2.2 For any two initial segments of well-orderings, if there is a correspondence between them that preserves the ordering, then that correspondence is unique.

Proof. Call the two initial segments A and B. Let us assume this is false, so that there is more than one order-preserving correspondence between A and B. Then there must be some element in one initial segment (let's say A) that can correspond to two different elements in the other, while the ordering is preserved. Because A is well-ordered, there must be some least element in A that corresponds to two different elements in B. Let's call this least element a, and let's call two of the elements it can correspond to in B b_1 and b_2, with $b_1 < b_2$. Because every element before a corresponds uniquely to some element in B, they must all correspond to elements less than b_1. So in the correspondence where a corresponds to b_2, there can be nothing that corresponds to b_1—everything less than a corresponds to something less than b_1, while everything greater than a must correspond to something greater than b_2. Thus, this second "correspondence" is not a correspondence at all, so there is a contradiction. □

(Alternatively, we could just notice that if A had two different order-preserving correspondences to B, then we could take one of those correspondences forwards, and the other backwards, and end up with an order-preserving correspondence from A to itself, which would contradict Theorem 2.1.)

Theorem 2.3 For any two well-orderings A and B, one of them has an order-preserving correspondence to an initial segment of the other (since every well-ordering is an initial segment of itself, this could mean that they correspond to each other).

Proof. Consider the set of order-preserving correspondences between initial segments of A and B. From the above result, we can see that any element in A that is in any such correspondence must correspond to the same element of B in all such correspondences. Thus, we can put all these correspondences in initial segments together to get a *maximal* correspondence between initial segments.

If every element of A is in one of these correspondences, this means that A corresponds to an initial segment of B. If every element of B is in one of these correspondences, this means that B corresponds to an initial segment of A.

So the only way the result could fail is if there are elements of both A and B that aren't in any initial segment with a correspondence.

If this happened, then since A and B are both well-ordered, we would be able to take the earliest non-corresponding element a of A and the earliest non-corresponding element b of B. Every element before a and b is corresponded by the maximal correspondence. But corresponding a and b to each other gets an even larger initial segment with a correspondence. This contradicts the maximality, so these must not both exist.

So one of the two well-orderings must correspond to an initial segment of the other. □

Ordinals are the names we use in these well-orderings, and now we have proven that there is just one sequence of ordinals—you can't have two different well-ordered sets where each has positions that don't occur in the other. So any two ordinals can themselves be ordered, based on which comes earlier in some well-ordering.

Theorem 2.4 For any non-empty set S of ordinals, there is a least ordinal in the set.

Proof. Let α be any element of S. If α is not the least ordinal in S, then consider any well-ordered set X where α occurs as the label of some position. Each lower ordinal in S must also be the label of some position in X. But since X is well-ordered, and the set of positions in X labeled with ordinals from S is non-empty, there must be some earliest position in X labeled with some ordinal from S. The label of this position must be the least member of S. □

Thus, the ordinals are themselves well-ordered.

Paradox 2.1 (Burali-Forti Paradox) Consider the set Ω of all ordinals. This set is well-ordered. If we put one more element at the end (or multiply it by 2, or by ω, or by any other ordinal) we would get an even larger well-ordered set. But this would mean there would have to be ordinals bigger than anything in Ω, even though Ω was the set of *all* ordinals!

2.7 Power sets

One more method we have of generating larger sets is the operation of taking the "power set". The *power set* of a set is the set of all subsets of a set. Here I list out all eight elements of the power set of $\{0, 1, 2\}$.

$$
\begin{array}{llllll}
\{ & & & & & \} = \emptyset \\
\{ & & & 2 & & \} \\
\{ & & 1 & & & \} \\
\{ & & 1, & 2 & & \} \\
\{ & 0 & & & & \} \\
\{ & 0, & & 2 & & \} \\
\{ & 0, & 1 & & & \} \\
\{ & 0, & 1, & 2 & & \}
\end{array}
$$

One of the things to notice—every time you add one more element to a set, you *double* the number of elements in the power set. Basically, the power set of the new set will contain all the subsets of the old set (none of which have the new element), and it will also contain one set corresponding to each of those subsets of the old set, with the new element added. For each subset of the old set, there are two subsets of the new set—one with and one without the new element.

Now we will show that even for infinite sets, the power set is a bigger set than the original set.

Theorem 2.5 (Cantor's Theorem) There is no correspondence between the set $\mathbb{N}$ of all natural numbers, and the power set of the natural numbers, $\mathcal{P}(\mathbb{N})$.

Proof. First, let's show this for the power set of the set of natural numbers. The way we will do this is by showing that no function f from $\mathbb{N}$ to $\mathcal{P}(\mathbb{N})$ can be a correspondence. In particular, we will show that no such function can have the onto property. So here is some particular function f from $\mathbb{N}$ to its power set:

$$
\begin{array}{lllllllll}
f(0) = & \{ & 0, & 1, & 2, & 3, & 4, & \ldots & \} \\
f(1) = & \{ & & & 2, & 3, & & \ldots & \} \\
f(2) = & \{ & 0, & & 2, & & 4, & \ldots & \} \\
f(3) = & \{ & 0, & 1, & & & 4, & \ldots & \} \\
f(4) = & \{ & & 1, & & 3, & & \ldots & \} \\
\vdots & & & & & \vdots & & & \\
\end{array}
$$

In order to show that this set is not onto, we need to find a particular element of $\mathcal{P}(\mathbb{N})$ that is not in the range of this function. We do this by going down the diagonal, and reversing everything, to generate this "diagonal' set D:

$$
\begin{array}{lllllllll}
f(0) = & \{ & \boxed{0,} & 1, & 2, & 3, & 4, & \ldots & \} \\
f(1) = & \{ & & \boxed{} & 2, & 3, & & \ldots & \} \\
f(2) = & \{ & 0, & & \boxed{2,} & & 4, & \ldots & \} \\
f(3) = & \{ & 0, & 1, & & \boxed{} & 4, & \ldots & \} \\
f(4) = & \{ & & 1, & & 3, & \boxed{} & \ldots & \} \\
\vdots & & & & & \vdots & & \ddots & \\
D = & \{ & & 1, & & 3, & 4, & \ldots & \}
\end{array}
$$

We can see that D is not in the range of f. If it were, then it would appear in some row of the table. But every row intersects the diagonal, and D has the *opposite* on the diagonal, making clear that it *can't* be the same as the set on any row of the diagram. □

Exercise 2.5 Show that this same argument works for *any* set, not just the set of natural numbers. That is, let f be a function from S to $\mathcal{P}(S)$, let $D = \{x \in S \mid x \notin f(x)\}$, and show that D isn't in the range of f.

It is straightforward to see that there is a correspondence between $\mathbb{N}$ and a *subset* of $\mathcal{P}(\mathbb{N})$—in particular, let $f(n) = \{n\}$, the set containing just that natural number. It is straightforward to see that this is a correspondence between $\mathbb{N}$ and the "singletons" (sets with just one element) of $\mathbb{N}$. (Again, this works for any set, not just $\mathbb{N}$.) Thus, we can see that the power set of any infinite set is in fact strictly larger than that set.

2.8 Some paradoxes

Paradox 2.2 Let V be the set of all sets. We can see that $\mathcal{P}(V)$ is the set of all sets of sets, and thus all its members are sets, so it is a subset of V. But Cantor's theorem says that $\mathcal{P}(V)$ cannot be in correspondence with any subset of V.

Note that the correspondence between $\mathcal{P}(V)$ and a subset of V is based on the identity function $f(x) = x$. Bertrand Russell plugged this into Cantor's proof and noted the more specific paradox.

Paradox 2.3 (Russell's Paradox) Let $D = \{x \in V \mid x \notin x\}$. That is, D is the set of all sets that are not members of themselves. Either D is a member of itself or it is not. If it is, then it does not satisfy the definition, so it should not be a member of itself. But if it is not, then it does satisfy the definition, so it should be. Either way, there is a contradiction.

To distinguish the sizes of infinity, we say that infinite sets that *can* be put in correspondence with the natural numbers are *countable* infinite sets, while the ones that can't are called *uncountable* infinite sets. However, it's important to note that although all countable infinite sets are the same size (because they can all be put in correspondence with each other), there are many different sizes of uncountable set, since the power set of one uncountable set is always an even larger uncountable set.

But this raises a natural question—is there any set whose size is somewhere in between that of $\mathbb{N}$ and $\mathcal{P}(\mathbb{N})$? Or in general, for any infinite set S, is there a set whose size is somewhere in between that of S and $\mathcal{P}(S)$? This turns out to be a deep and difficult question.

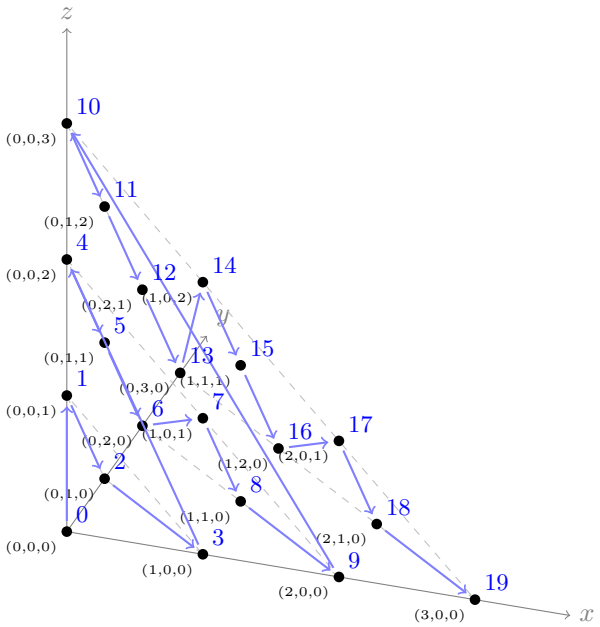
3 Many countably infinite sets

Last week introduced a lot of new ideas. This week will not involve as many new concepts—we will just be working with lots of infinite sets.

We have already shown that the set $\mathbb{N}\times\mathbb{N}$ of ordered pairs of natural numbers can be put in correspondence with the set $\mathbb{N}$ of natural numbers, and in fact we have shown this in two ways. One way involved writing them out in a grid, and numbering them in zigzag fashion through the grid, so that every pair was assigned a natural number. The other way involved assigning the odd numbers to the pairs beginning with 0 (leaving the infinitely many even numbers in reserve), and then assigning every other even number to the pairs beginning with 1 (leaving the infinitely many multiples of 4 in reserve), and then assigning every other multiple of 4 to the pairs beginning with 2 (leaving the infinitely many multiples of 8 in reserve), and so on. On this method, the pair (m, n) is represented by the natural number $2^m \cdot (2n + 1)$. It's clear that each pair is represented by a distinct natural number.

These methods naturally generalize to the ordered triples of natural numbers. To do Cantor's method, we have to write the triples in a three-dimensional array, and then zigzag through triangular slices:

Cantor-style Enumeration of $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$



Each triple (x, y, z) gets a unique natural number following the blue path

To generalize Hilbert's method, instead of sending the pair (m, n) to the number 2^m multiplied by the nth odd number, we send the ordered triple (m, n, ℓ) to the number 2^m multiplied by 3^n multiplied by the ℓth number that is neither divisible by 2 nor by 3.

But we can also do something more general. Note that we can think of an ordered triple (m, n, ℓ) by grouping the first two terms, $((m, n), \ell)$. We already know there is a correspondence between the ordered pairs (m, n) and the natural numbers. (In fact, there are many.) If the pair (m, n) corresponds to k under this ordering (we don't have to worry about *which* correspondence we use, just that it is in fact a correspondence), then we can send the triple (m, n, ℓ) to the pair (k, ℓ), and we can check that this is a correspondence between

the ordered triples and the ordered pairs. But then since the ordered pairs correspond to the natural numbers, we can put these two correspondences together to get a correspondence between the ordered triples and the natural numbers.

This last method clearly generalizes to the ordered quadruples, and ordered quintuples, and so on.

Now we have some exercises about proving that various sets are countably infinite. Recall that what that specifically means is that we need to show that these sets have a correspondence with the set of all natural numbers. For the first one, we need to figure out how to fix a correspondence that leaves out some of the natural numbers, and for the others we need to put together some of the results we've already shown.

Exercise 3.1 Any infinite subset of a countably infinite set is countably infinite.

Exercise 3.2 The union of two countably infinite sets is countably infinite.

Exercise 3.3 The union of countably many sets that are each countably infinite is countably infinite.

3.1 Rational numbers

We have talked about two sets of numbers: $\mathbb{N} = \{0, 1, 2, \dots\}$, the set of natural numbers, and $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of all integers. The natural numbers are used for counting, and saying how many members there are in a finite set. The negative integers are actually a bit harder to explain any use for, but it's easy to see how to define them (or at least, to define names for them—just put a minus sign in front of the name of a non-zero natural number).

One thing that many ancient people noticed is that *ratios* of numbers are also important. If one pipe is twice as long as another, then the musical note made by blowing into the shorter one is exactly an octave (a particularly simple sound) above the musical note made by blowing into the other. And if the pipes are in other ratios, you get other musical intervals—a ratio of 3:2 gives the interval of a perfect fifth (from C up to G), and a ratio of 5:4 gives the interval of a major third (from C up to E), both of which are particularly pleasing. But if the ratio of the pipes is not so simple, then the interval is not so pleasing.

Similarly, if you are looking upwards at a particular angle at the top of a tree or a building or a mountain, the angle you are looking at doesn't precisely tell you how tall the tree or building or mountain is—but it *does* tell you the *ratio* of the height of the object to its distance from you.

Both of these ideas involve measurement of some length (or distance or height), in terms of another one as a unit. The followers of Pythagoras initially thought that all of the universe could be measured through these ratios, and so they came up with a larger set of numbers, called the "rational numbers". In modern notation, we write rational numbers as fractions (or "quotients"), like $2/3, 4/6, 3/2, 6/4$. But we notice that some of those fractions actually represent the same ratio—so we say $2/3 = 4/6$ and $3/2 = 6/4$. The rational numbers are named as $\mathbb{Q}$ for "quotient". It's clear that there is a correspondence between the rational numbers and some subset of the set of ordered pairs of integers. Thus, the set of rational numbers is also countably infinite.

But as we will see, the Pythagoreans were wrong—there are distances that can't be measured with any ratio, no matter what unit you are using.

3.2 Sequences, sets, sentences

For the next one, we need to consider the set of all finite sequences of natural numbers. A finite sequence is either a single natural number, or an ordered pair of them, or an ordered triple of them, or ..., or an ordered n-tuple, for some natural number n.

Exercise 3.4 The set of all finite sequences of natural numbers is countably infinite.

Remember that the difference between a finite *sequence* and a finite *set* is that the elements in a sequence have an order (and can repeat) while the elements of a set don't. For any sequence of natural numbers, we can associate it with the set of natural numbers that occur at any position in this sequence. This associates many sequences (in fact, infinitely many) with each set of natural numbers. (Why?) But if we just restrict

ourselves to the subset of finite sequences whose elements are listed in increasing order, this association gives a correspondence with the set of all finite sets of natural numbers. (Why?) Thus, the set of all finite sets of natural numbers has a correspondence to an infinite subset of a countably infinite set, so it is also countably infinite.

At some point next week we will also be interested in the concept of a *co-finite* set of natural numbers. That means that the complement of the set (i.e., the set of all natural numbers that *aren't* in it) is finite. That is, after some point, all the remaining natural numbers are in the set. There is a natural correspondence between finite sets and co-finite sets (just take the complement of the set), so the set of co-finite sets of natural numbers is also countably infiite.

Exercise 3.5 The set of all grammatical sentences in English is countably infinite.

Exercise 3.6 The set of all syntactically correct computer programs is countably infinite.

Exercise 3.7 The set of all definitions of sets of natural numbers is countably infinite.

Paradox 3.1 (Richard's Paradox) Consider the set of all definable sets of natural numbers. This set must be countably infinite, because we just argued that the set of definitions of sets of natural numbers is countably infinite. So this set can be enumerated with the natural numbers.

Now let us use the diagonal argument to define a set that is not on that list. That is, define the set R by saying $n \in R$ iff n is *not* a member of the set of natural numbers defined by the nth definition in our list of definitions of sets of natural number. R can't be the same set as any set in our list, because if it were the same as the set at the nth position in the list, then it would have n iff it *didn't* have n.

But we just defined R, and we said that the list contains all the definitions of sets of natural numbers.

4 Comparing the real numbers and the power set of the natural numbers

4.1 Measurement of extensive quantities

So far we have talked about the natural numbers $\mathbb{N} = \{0, 1, 2, \dots\}$ for counting how many of something there are, the integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, and the rationals $\mathbb{Q} = \{1/2, -3/4, 7/5, -137/42, \dots\}$ for measuring ratios of these. But there is one more important set of numbers, known as the real numbers, $\mathbb{R}$.

These are the numbers we use for measuring quantities like distance, time, volume, and mass. (I am specifically leaving out temperature for now—we'll see some important reasons why it is different later. The quantities discussed here are often called "extensive quantities", and they are said to be measured on a "ratio scale". Most of the ideas in this section derive from the 1971 book *Foundations of Measurement*, by Krantz, Luce, Suppes, and Tversky, and related work by the individual authors.) As we will show, it turns out that $\mathbb{Q}$ is not enough for measuring these, and we need something else.

Let's think carefully about what is involved in measuring these, *without* using any concepts of number. First, we need a concept of when one thing is larger, smaller, or equal to another. It is simplest to understand this for distance or mass. To compare two distances, we have to imagine stretching a stick, or a taut string, between each pair of points whose distance we are comparing, and then dragging these sticks or strings around until they are next to each other, and seeing which sticks out past the end of the other. To compare two masses, we have to imagine taking the two objects, and putting them on a balance, and seeing which direction the balance tilts. Obviously, we can't actually engage in this physical process for all distances or all masses, but we believe it is possible.[1]

We can represent this relation with the symbols ">" and "<". Importantly, both of these relations are orderings, as we have defined before—they are irreflexive, anti-symmetric, and transitive. Nothing is longer than itself; if x is longer than y, then y is not longer than x; and if x is longer than y and y is longer than z, then x is longer than z. These are empirical facts about how these concepts behave, that didn't have to be the case. It doesn't hold for things like "beats at chess" (you could imagine three players that have a rock-paper-scissors relation to each other). It didn't even have to be the case that dragging things around like this led to *any* sort of meaningful comparison—you could imagine that there would be some objects such that if you measure them in one room the first is longer, while if you measure them in another room, the second is. (This sort of thing clearly happens with beating at chess.) But we do in fact get a *linear* order for each of these quantities, defined by the trichotomy principle: for any x, y, either $x < y$ or $x > y$ or $x = y$.

The important defining feature for extensive quantities (as opposed to other types of quantities like temperature) is that there is also *another* feature that is relevant to their measurement. You can "concatenate" two quantities of the relevant sort to produce a third quantity of the relevant sort: you can stick two rods together end-to-end, start one duration right when another ends, or stick two objects together for volume and mass. We represent this concatenation with the symbol "$\circ$". The first properties we note for concatenation are that $x \circ y = y \circ x$ (commutativity) and $x \circ (y \circ z) = (x \circ y) \circ z$ (associativity). It doesn't matter which order you stick things together, any more than it matters which direction they are pointing when you compare their length or duration or volume or mass.

[1] I won't talk about how to measure time—I'm not sure how you would drag two separate temporal intervals around until one of them starts at the same time as the other, so the only ways I know to measure time do presuppose numbers. Measurement of volume was the famous historical "eureka" moment. Someone had gifted the king a crown and claimed it was made of pure gold, and the king tasked Archimedes with verifying that this was true. Archimedes knew that gold would have more mass than any other substance he knew at a given volume. But he didn't know how to measure the volume of the crown without pounding it into a cube, which of course the king didn't want. The story goes that one day, when Archimedes got into a completely full bath, and he noticed some water spill out of the tub, he realized he had found his solution, and shouted "ηυρηκα" ("eureka", "I have found it") and ran out of the bath, still naked, to go tell the king. The idea was that if you have a completely full tub of water, and submerge one object in the water, an amount of water will spill equal to its volume. If you then submerge the other object, you can see whether this causes even more water to spill, or leaves the tub not quite full.

4.2 Axioms for extensive measurement

The important features extensive quantities have in common are the interactions between concatenation and comparison. It's worthwhile to think about each of the different extensive quantities I have mentioned to have an intuitive understanding of why each of the axioms I will list holds for them.

For any x and y, $x \circ y > x$.

If $x > y$, then for any z, $x \circ z > y \circ z$.

Differences: If $x > y$, then there is some z with $x = y \circ z$

This axiom of differences effectively says that there is always a length/duration/volume/mass equal to the leftover amount whenever one is larger than the other, which we can think of as the "difference" between x and y. These axioms allow us to show that differences are unique—that is, there can't be two different z and z' with $x \circ z = x \circ z'$. If z and z' were different, then one would have to be greater than the other, say $z > z'$, but then the second axiom above would show that $x \circ z > x \circ z'$, and by irreflexivity of the ordering, $x \circ z \neq x \circ z'$.

Exercise 4.1 Show, using just these axioms, and the ones for ordering and concatenation separately, that if $x_1 \circ y_1 = x_2 \circ y_2$, and $x_1 < x_2$, then $y_1 > y_2$. (Hint: If it weren't the case that $y_1 > y_2$, what would go wrong?)

There is one more, particularly important, finite axiom:

Divisibility: For any x there is an x_2 with $x_2 \circ x_2 = x$.

We think of x_2 as "half" of x. By applying the same process to x_2, it's easy to see there must be x_4 with $x_4 \circ x_4 \circ x_4 \circ x_4 = x$, and similarly with division into 8, 16, 32, etc. equal pieces. It turns out that later axioms will entail that there is also an x_3 with $x_3 \circ x_3 \circ x_3 = x$, and an x_5 with $x_5 \circ x_5 \circ x_5 \circ x_5 \circ x_5 = x$, and so on. But we can think of divisibility either strictly as division into two equal pieces, or generally as division into arbitrary finite collections of equal-sized pieces.

These properties we have mentioned so far give rise to the basic familiar properties of measurement. If we want to measure the volume of a big vessel, we can take some container and designate it as our unit and call it a "cup", and see how many cups we have to "concatenate" (i.e., pour the volume of the cup into the vessel) to fill it. If we can't fill it exactly with some number of cups, then we use divisibility to find a smaller unit, say an "ounce" that needs to be concatenated 8 times to fill a cup, and concatenate these ounces to try to fill the vessel. If that doesn't precisely fill it either, then we get an even smaller unit, like a "tablespoon" that needs 2 copies to fill the ounce, or a "teaspoon" that needs 3 copies to fill the tablespoon. There's no particular reason to use one set of units rather than another, so in some measurement systems, we do it systematically and divide by 10 each time, but traditionally people used whatever units were convenient.

If this above process terminates in a finite number of steps, then the vessel can be measured as some rational multiple of the "cup" that we used as the unit. But there are two more principles we need to deal with cases where this doesn't terminate. These are known as the "Archimedean principle" and the "Continuity principle".

The Archimedean principle rules out a failure at the first step of the above process. It says that for any two quantities of the type we are measuring, there is some *finite* number of copies of the smaller one that concatenate to give a larger quantity than the other. More explicitly:

For any x, y, either $x < y$ or $x < y \circ y$ or $x < y \circ y \circ y$ or $x < y \circ y \circ y \circ y$ or ....

This rules out the possibility of a failure where we have a giant vessel, and keep pouring water into it a cup at a time, but no matter how many cups we have poured, we can keep going. That is, it rules out the possibility of one quantity being infinitely larger than another. It ensures that our vessel can be approximated in volume to within one cup by some natural number of cups. We could imagine non-Archimedean quantities,

where there really are infinitely different scales, and mathematicians study such systems, but in ordinary measurement of distance, time, volume, and mass we assume they are Archimedean.

The Continuity principle considers what happens if the measurement process never terminates. There's no guarantee that every volume will be some specific natural number of cups, or some natural number of ounces, or some natural number of tablespoons, or some natural number of teaspoons. In fact, by divisibility, for whatever unit we start with, we can always find a smaller one, and concatenate any number of copies of it to produce quantities that can't be measured with a natural number of the original unit. What Continuity says is that even if we repeat this process *infinitely* often, as long as we do so with smaller and smaller units, so that there is *some* upper bound that it never passes, then there is a quantity that the sequence will converge to. More precisely:

> If $x_0 < x_1 < x_2 < \cdots < y$, then there is some L such that $x_0 < x_1 < x_2 < \cdots < L$, and there is no z with $x_0 < x_1 < x_2 < \cdots < z < L$.

Note that the principle also means that any bounded *decreasing* sequence has a limit. If there is a bounded decreasing sequence $x_0 > x_1 > x_2 > \cdots > y$, then by letting z be any quantity greater than x_0, the axiom above of differences tells us there is x_0' with $x_0 \circ x_0' = z$, and x_1' with $x_1 \circ x_1' = z$, and x_2' with $x_2 \circ x_2' = z$, ..., and y' with $y \circ y' = z$. Based on our above result that taking differences reverses ordering, we see that $x_0' < x_1' < x_2' < \cdots < y'$ gives us a bounded increasing sequence, which must have a limit L'. And we can check that the value L with $L \circ L' = z$ will be the limit of the x_i.

To get the full sense of what Continuity actually means, it is helpful to think about what it would take for it to be false. There would have to be some infinite, increasing sequence $x_0 < x_1 < x_2 < \cdots < y$ that *didn't* have a limit. To see what this means, let's call $y = y_0$. If there is no limit for the sequence, then y_0 is not itself the limit, so there is some $y_1 < y_0$ with y_1 still greater than every x_i. But since y_1 is also not the limit, there would have to be $y_2 < y_1$ that is still greater than every x_i. What Continuity rules out is the possibility of having $x_0 < x_1 < x_2 < \cdots < y_2 < y_1 < y_0$ without having *anything* in the middle. That is, it says that for any coherent sequence of approximations you can have from above and below, there must be some quantity that would be in between all of them. There are no gaps in the sequence of potential quantities.

4.3 Decimal notation for quantities

Now that we have listed the substantive assumptions about measurement for extensive quantities, we can show that all the extensive quantities are basically the same. We will define the real numbers $\mathbb{R}$ as the structure these extensive quantities all share.

Measurement of any extensive quantity involves arbitrary selection of a unit. So for instance, you might take the length of your foot as a unit, and (if your feet are the same length) then roughly measure a distance by counting how many times you can put one foot immediately in front of the other to get up to that distance. By the Archimedean principle, no matter what unit you choose, and what distance you are trying to measure, there will be a natural number that is this rough approximation. If the unit is your foot, and you are measuring your friend's height (they probably have to lie down, so that you can get your feet lined up with them) it will probably be about 5 or 6. But if you went for a run this morning and try to measure the distance of your run in feet, you might end up with a number more like 17,384 (and it'll take you a lot longer to actually do this measurement than it took you to do the run). Even if you wanted to measure the distance from Los Angeles to New York, the Archimedean principle tells us that *some* number of feet will do it, but it'll probably take too long to actually do this measurement. If you concatenate two distances, the number of units in the concatenated distance will be approximately the sum of the numbers in each (possibly off by one, depending on the rough leftovers).

If you want to get a more precise measurement, then you can use divisibility to find a smaller unit that goes into the first one some precise number of times. Maybe the first joint of your thumb is exactly the right size for twelve copies of it to go into the length of your foot, and you can call this an "inch". This might allow you to measure your friend's height more precisely. Perhaps it is basically five copies of your

foot concatenated with four copies of your thumb. You can't represent this with a single number, but if you have a symbol for each unit, then you can make it clear, like $5'4''$.

If you're doing some kind of specialized measurement where you really need details, you might want more than just a single subdivision. For instance, in antiquity, most people were fine with time being measured in hours, which could be defined as the length of time it takes for the sand in a particular glass to fall to the bottom. But astronomers often wanted very precise measurements to say exactly when various stars rose over the horizon on particular dates, and when precisely a planet crossed over from one constellation into the next. So Ptolemy defined the "*pars minuta prima*" ("first small part") as 1/60 of an hour, and the "*pars secunda minuta*" ("second small part") as 1/60 of that unit. This allowed him to say things defining the coordinate of a star in the sky based on precisely when it would rise on a particular day of the year, like saying that Sirius would rise at 06:45:08. It wasn't until thousands of years later that anyone other than astronomers cared about measuring time this precisely, so they borrowed these Latin words "minute" and "second" for the smaller units.

But based on Divisibility and Continuity, we have assumed that for any finite set of units, there are quantities that can't be precisely measured with them, and in the modern world we want to have a systematic way to talk more precisely about these quantities. So rather than doing whatever division into smaller parts is convenient based on some actual object on hand (like your thumb, or a particular smaller glass of sand), we use Divisibility to systematically divide each unit into a smaller part in the *same* way every time. Because there are exactly ten natural numbers we can represent with a single symbol (0, 1, 2, 3, 4, 5, 6, 7, 8, 9), it has become standard to divide into ten equal parts at each next step.

We still have to start with one arbitrary unit, but we can report the result of a precise measurement by putting a dot after the count of that base unit, and then represent the quantities of smaller units by a sequence of digits after this dot. This is the familiar idea we have for representing a distance as 1609.34 meters (1609 meters, concatenated with 3 units that are a tenth of a meter, concatenated with 4 units that are a tenth of a tenth of a meter), or a duration as 9.58 seconds (9 seconds, concatenated with 5 units that are a tenth of a second, concatenated with 8 units that are a tenth of a tenth of a second), or a volume as 3.78541 liters, or a mass as 453.592 grams. This notation gives us a way of representing any quantity that can be built by concatenating some precise natural number of copies of our base unit, and tenths of that base unit, and tenths of that unit, up to some finite number of subdivisions into tenths. Because the Latin word for "tenth" is "*decimus*", we call the dot the "decimal point", and call this notation "decimal notation".

Theorem 4.1 Every quantity has a representation in decimal notation if we allow an *infinite* sequence of digits.

Proof. To represent a quantity x, note that by the Archimedean principle, some finite number of copies of the base unit must concatenate to more than the base unit, so there must be some specific finite number n of copies concatenate to no more than x, while $n+1$ are at least x. That is $n \leq x \leq n+1$. This gives a first approximation to x.

Since we know that n copies of the base unit are no more than x, and $n+1$ copies of the base unit are at least x, we know that n copies of the base unit and 10 copies of the tenth part add up to at least x, so there must be some $d_1 \in \{0,1,2,3,4,5,6,7,8,9\}$ such that n copies of the base unit and d_1 copies of the tenth of this unit add up to no more than x, while n base units concatenated with d_1+1 copies of the tenth part add up to at least x. That is, $n.d_1 \leq x \leq n.d_1 + 0.1$. (I assume you know how to add 0.1 in decimal notation, even though I haven't defined how to do it. Note: I can't just write "$n.(d_1+1)$", because d_1 might be 9, and then d_1+1 wouldn't be a digit, so this wouldn't be a proper decimal notation. Thus, you have to do addition with your familiar process of carrying the 1 if needed.) $n.d_1$ is then a second approximation to x.

Similarly, there must be some $d_2 \in \{0,1,2,3,4,5,6,7,8,9\}$ such that $n.d_1d_2 \leq x \leq n.d_1d_2 + 0.01$, making $n.d_1d_2$ a third approximation to x. And again, there is $d_3 \in \{0,1,2,3,4,5,6,7,8,9\}$ such that $n.d_1d_2d_3 \leq x \leq n.d_1d_2d_3 + 0.001$, so that $n.d_1d_2d_3$ is a fourth approximation, and so on. We can always continue this process, and this guarantees that every quantity has a representation with an infinite decimal notation (which could involve infinitely many 0's, if one of the finite approximations was precisely accurate). □

We have shown that every quantity has *at least* one infinite decimal representation. Now we will show

that every infinite decimal representation has *exactly* one quantity that it represents.

Define $[x, y]$ to be the set $\{z \mid x \leq z \leq y\}$ of all quantities between x and y, including the endpoints. Then $[n.d_1d_2\ldots d_n, n.d_1d_2\ldots d_n + .00\ldots1]$ is the set of all quantities that share the same $(n+1)$th approximation.

Exercise 4.2 Consider the sequence of nested intervals $[n, n+1]$, $[n.d_1, n.d_1+.1]$, $[n.d_1d_2, n.d_1d_2+.01]$, ... This is the set of all quantities that share a given $(n+1)$th approximation. Use the Continuity principle to show that there must be *at least* one quantity contained in the intersection of all these intervals, and use the Archimedean axiom to show that there must be *at most* one quantity contained in the intersection of all these intervals. (Hint: If there were two distinct quantities contained in this intersection, see what wold happen if you count how many copies of their difference go into the original unit.)

We have shown that for every quantity, there is at least one infinite decimal representation, and for every infinite decimal representation, there is at least one and at most one quantity. Thus, we can think about quantities by thinking about these infinite decimals, which is what we usually do. If we could show that for every quantity, there is at most one decimal representation, then there would be a perfect correspondence between quantities and infinite decimal representations.

But unfortunately, this is false.

Consider what happens when $n = 0$ and every $d_i = 9$. Then we are considering the sequence of intervals $[0, 1]$, $[0.9, 0.9 + 0.1]$, $[0.99, 0.99 + 0.01]$, $[0.999, 0.999 + 0.001]$, ... This is the sequence of intervals $[0, 1]$, $[0.9, 1.0]$, $[0.99, 1.00]$, $[0.999, 1.000]$, $[0.9999, 1.0000]$, ... We can see that the infinite decimal $0.999\ldots$ must represent a quantity inside every single one of these intervals. But we can see that the infinite decimal $1.000\ldots$ *also* represents a quantity inside every single one of these intervals. Since we have proven above that there can be at most one quantity contained in the intersection of all these intervals, this means these two infinite decimals must represent the *same* quantity! See Vi Hart's excellent video for more explanation of this fact: `https://vimeo.com/147787605`.

However, it turns out that the *only* infinite decimals that represent the same quantities are the ones of the form $n.d_1d_2\ldots d_i999\ldots$ and $d.d_1d_2\ldots(d_i+1)000\ldots$, where n is any natural number, and the d's are any digit, except that d_i is not 9. It's a bit fussy to check this, but basically, if you take any infinite decimal and truncate it to a finite one, then you can see that the truncation is *strictly* less than the infinite decimal as long as the part you truncated wasn't infinitely many 0's. Similarly, if you take any infinite decimal and truncate it to a finite one, and add 1 to the last digit (with whatever carrying is needed), then if the part you truncated wasn't infinitely many 9's, then the result must be strictly greater than the infinite decimal you started with. Thus, for any two infinite decimals that don't end with infinitely many 9's or infinitely many 0's, by taking the first digit at which they differ, you can find a quantity that is strictly above one and strictly below the other, and thus show that the infinite decimals must correspond to distinct quantities.

This basically corresponds to our intuitive idea that you can tell which of two decimal notations is greater by finding the first digit at which they differ, and seeing which is larger there, with the specific exception involving infinitely many 0's or 9's.

4.4 Operations on decimal notations

We have now shown that every decimal notation corresponds to a quantity, and every quantity has either exactly one or exactly two decimal notations. It is useful to think about how these decimal notations interact with the axioms of extensive measurement.

First, we have to see how concatenation works. If we concatenate two quantities that are represented by *finite* decimal notations, it is straightforward to see that the standard algorithm we learned in school for adding decimals will work (that is, you add the digits one by one, with a tricky bit about carrying the 1 if two of the digits add up to ten or more, or add up to 9 and have a 1 carried from the place below).

If the quantities are represented by *infinite* decimals, then it's a bit trickier. We can truncate each infinite decimal at a finite number of digits to get notations like $n.d_1d_2\ldots d_i$ and realize that $n.d_1d_2\ldots d_i$ is no more than the number, while $n.d_1d_2\ldots d_i + 0.00\ldots1$ is no less than the number. By the first few axioms of extensive measurement (including commutativity of concatenation, and transitivity of comparison), we

know that if $x_\ell \leq x \leq x_u$ and $y_\ell \leq y \leq y_u$, then $x_\ell \circ y_\ell \leq x \circ y \leq x_u \circ y_u$. Thus, we can produce a notation for the concatenation of two quantities whose notations we know by using these truncations. But it can sometimes be hard to be sure exactly what this notation is going to look like if a lot of the digits happen to add up to 9 in a row (since somewhere down the line, the digits might add up to more than 9, and trigger a carry that goes way back to an earlier position, or might add up to less than 9, and settle that the previous ones will definitely be 9's with no carry).

Reviewing the other axioms from above, we can see that two of those axioms state the existence of further quantities. The axiom of differences stated that if $x > y$, then there is some z with $x = y \circ z$. The result at the end of section 4.3 showed that if $x > y$, then at the first point where the decimal notations for x and y differ, x has a greater digit than y. Then the standard algorithm we learned in school for subtracting decimals will work.

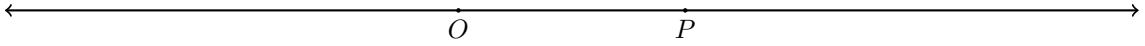
The axiom of divisibility is somewhat trickier. This stated that for any x, and any natural number n, there is an x_n such that x_n concatenated with itself n times yields x. If we have a notation for x, then it turns out that the standard "long division" algorithm we learned in school will allow us to produce a notation for x_n.

To see this, think about how we can find a notation for the number that, added to itself four times, yields $\pi = 3.1415926\ldots$. If we call this number $\pi/4$, then we can see that first, $0 < \pi/4 < 1$, because $0 + 0 + 0 + 0 = 0 < \pi < 4 = 1 + 1 + 1 + 1$. Then we can see that $0.7 < \pi/4 < 0.8$, because $0.7 + 0.7 + 0.7 + 0.7 = 2.8 < \pi < 3.2 = 0.8 + 0.8 + 0.8 + 0.8$. Then we can see that $0.78 < \pi/4 < 0.79$ because $0.78 + 0.78 + 0.78 + 0.78 = 3.12 < \pi < 3.16 = 0.79 + 0.79 + 0.79 + 0.79$.

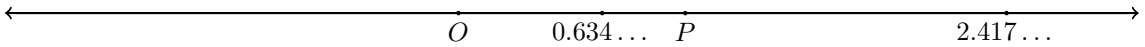
At each point, we find the next digit by taking the lower bound we got at the previous step and multiplying by 4, subtracting this off from the notation we have, finding the digits of what is left, and finding out what digit at the next step would go 4 times into what's left. In fact, this construction allows us to show that even if we had only assumed divisibility into ten equal pieces, Continuity and the Archimedean principle would allow us to construct these notations, and that would be enough to ensure divisibility into any natural number of equal pieces. In section 4.6 we will show the equivalent thing for any other single instance of divisibility, so it's standard to only assume divisibility into two equal pieces.

4.5 The real line, and an interval

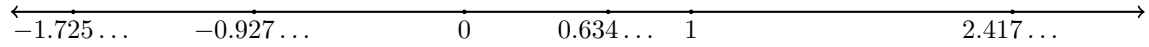
The above considerations show that every extensive quantity can be represented by an infinite decimal in an almost unique way. At this point, we will now specifically focus on the quantity of distance. Let us think about the points on an infinitely long line, and pick one of these points we will call "O" to be the "origin", and some other point we will call "P".



If we use the distance between O and P as the unit, then we can represent any point between O and P with an infinite decimal between 0 and 1, and any point beyond P with an infinite decimal greater than 1. In each case, we will use the infinite decimal corresponding to the distance to the point O. It should be clear that for every point on this line, there is a distance between that point and O, and for every distance, there is exactly one point with that distance from O in the direction of P.



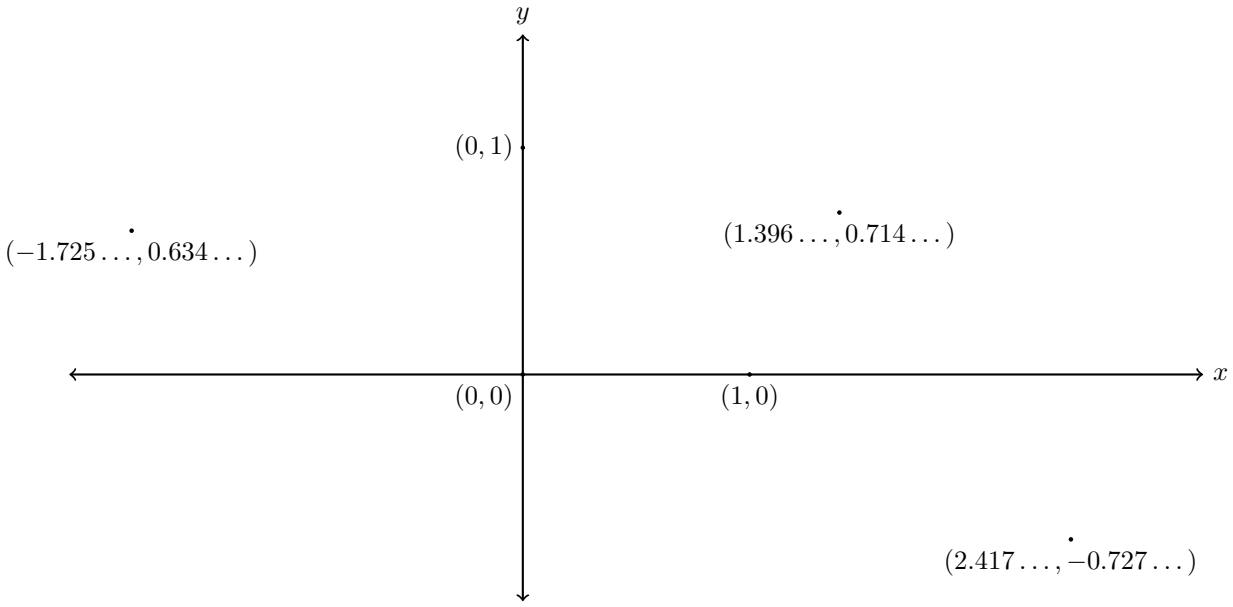
For every distance, there is of course also exactly one point with that distance from O in the direction *opposite* P, and we can represent this point with an infinite decimal with a minus sign in front, and represent O itself with "0".



This is what we think of as the "real line", and we can think of the real numbers $\mathbb{R}$ as corresponding exactly to the positions on this line. (Some real numbers have two different decimal representations, like

$0.999\cdots = 1.000\ldots$, but this is no more problematic than some rational numbers having two different fractional representations, like $1/2 = 3/6$. The numbers are the shared thing that the notations represent.) In the cases of distances and times, we can sometimes use negative real numbers to represent opposite directions, but it's usually harder to make sense of this for volumes and masses—actual extensive quantities are represented with positive real numbers. And it turns out that for other quantities that don't have a similar concept of concatenation, if there is some way to measure them (as people often claim for things like temperature, density, monetary value, moral goodness, redness, IQ, personality type, etc.), usually the real numbers will work, but we often need the negative real numbers as well as the positive ones.

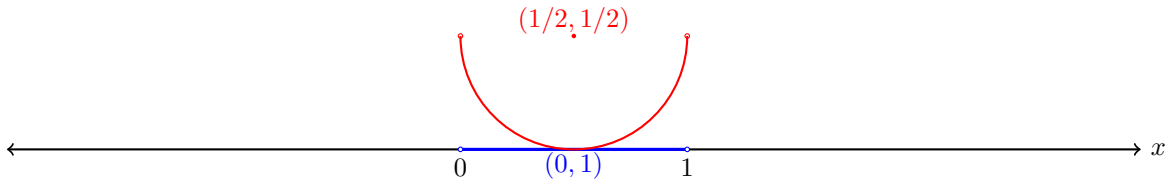
René Descartes noticed that in a plane, there is not just a unique concept of distance between two points, but also a concept of distance between a point and a line. If we draw two perpendicular lines in the plane, we can then represent each point uniquely by the ordered pair of real numbers, indicating the distance (positive or negative) between the point and each of these two lines, which we call the x- and y-axis. This should of course be familiar from geometry classes.



We can then use ideas from geometry in the plane to show that there are correspondences between the set of all real numbers (which we think of as the full line) and various subsets. (Just as there were correspondences between the set of all natural numbers, and various subsets.)

For these exercises, it is useful to remember that for any two points, there is exactly one line going through both of them, and for any two lines, if they are not parallel, then there is exactly one point at which they intersect. Furthermore, for any circle, there is a unique point called its "center", and every line through the center of the circle intersects the circle exactly twice, and the two points at which they intersect are in opposite semicircles.

Exercise 4.3 Show that there is a correspondence between the set $\{x \in \mathbb{R} \mid 0 < x < 1\}$ (all real numbers between 0 and 1, blue in the diagram below) and the set of points on the semicircle of radius $1/2$, centered at $(1/2, 1/2)$ (red in the diagram below), not including the endpoints of either the interval or the semicircle. Make sure to show that all four properties of a correspondence hold.



Exercise 4.4 Show that there is a correspondence between the set of points on the semicircle of radius 1/2, centered at $(1/2, 1/2)$ (not including the endpoints) and the set of *all* real numbers. Make sure to show that all four properties of a correspondence hold.

Exercise 4.5 Show that there is a correspondence between the set of points in any particular finite interval of real numbers (not including endpoints) and the set of *all* real numbers.

4.6 Other base notation

In section 4.3 we established a notation system for the real numbers with infinite decimals, by taking any arbitrary unit, and dividing it into sub-units that are each a tenth of the previous unit. Obviously, using the Divisibility axiom, we could do the same thing with sub-units constructed in any other systematic way.

In decimal notation, we have ten basic symbols (0, 1, 2, 3, 4, 5, 6, 7, 8, 9) and we use them to stand for the first ten natural numbers. We represent larger natural numbers by saying that the symbol in each position to the left of the first stands for how many copies of a number ten times as large we are adding. Thus, we have the tens place, the hundreds place, the thousands place, and so on. When we extended this to the real numbers, we weren't *counting* any more, but were *measuring*, and we had each place to the right after the dot stand for units that were one tenth of the previous unit, so we had the tenths place, the hundredths place, the thousandths place, and so on.

There is nothing particularly mathematically special about using ten here—it's just that before we used symbols, or even words, to count, most societies used body parts, and the most easily manipulated body parts for counting happen to come in ten. Thus the words for counting repeat in patterns of ten, and this made it easiest to use a notation system of this type based on ten symbols.

However, we can do the same thing with any other finite number of symbols, as long as we agree on what those symbols are. I will indicate these alternate notations by replacing the ordinary dot '.' with a dot with a subscript (written in decimal notation!) for the number of symbols we are using. Thus, I will use '$._8$' for "octal" notation using the eight symbols 0, 1, 2, 3, 4, 5, 6, 7, and '$._2$' for "binary" notation using the two symbols 0, 1. I will even use '$._{16}$' for "hexadecimal" notation using the sixteen symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F. The number that I subscript the dot with (the number of distinct symbols available for each position) is called the "base" of this notation.

If I want to represent the number 108.25 in any of these other bases, I can do it as follows.

To represent it in octal (base 8) notation, first, I note that the powers of 8 are $\ldots, 64, 8, 1, 1/8 = 0.125, \ldots$. To represent 108.25 with these, I first note that $1 \cdot 64 = 64 < 108.25 < 128 = 2 \cdot 64$, so the first digit must be 1, so the notation will be $1??._8?? \ldots$. What's left over is 44.25, and $5 \cdot 8 = 40 < 44.25 < 48 = 6 \cdot 8$, so the next digit must be 5, so the notation will be $15?._8?? \ldots$. What's left over is 4.25, and $4 \cdot 1 = 4 < 4.25 < 5 = 5 \cdot 1$, so the next digit must be 5, so the notation will be $155._8?? \ldots$. What's left over is 0.25, and $4 \cdot 0.125 = 0.25$, so the notation must be precisely $155._84$.

To represent it in binary (base 2) notation, we note that the powers of 2 are $\ldots 128, 64, 32, 16, 8, 4, 2, 1, 1/2 = 0.5, 1/4 = 0.25 \ldots$. I'll go through it more quickly, and note that $108.25 = 64 + 32 + 8 + 4 + 0.25$, so the notation must be precisely $1101100._201$.

To represent it in hexadecimal (base 16) notation, we note that the powers of 16 are $\ldots 256, 16, 1, 1/16 = 0.0625, \ldots$. Note that $6 \cdot 16 = 96 < 108.25 < 112 = 7 \cdot 16$, so the first digit must be 6, so the notation will be $6?._{16}?? \ldots$. What's left over is 12.25, and $12 \cdot 1 = 12 < 12.25 < 13 = 13 \cdot 1$, so the next digit must be the 12th symbol in our set, which is C, so the notation must be $6C._{16}?? \ldots$. What's left over is 0.25, and $4 \cdot 0.625 = 0.25$, so the notation must be precisely $6C._{16}4$.

These notations happen to be even more convenient for computers to use than decimal (base 10) notation is for humans. Hexadecimal happens to be used for a lot of things, particularly for certain representations of colors. Something like #FFA753 represents the color with FF._{16}=255 brightness in red pixels, A7._{16}=167 brightness in green pixels, and 53._{16}=83 brightness in blue pixels, which works out to a particular shade of bright orange, `https://www.color-hex.com/color/ffa753`.

In section 4.5 we proved that there is a correspondence between the real numbers in an interval, and the set of all real numbers, and the next two exercises attempt to show there is a correspondence between the real numbers in an interval and the power set of the natural numbers. However, all we can directly prove is correspondences between each of these sets and a subset of the other. Using base notation other than 10 happens to be helpful for these.

Exercise 4.6 Using binary notation, show that there is a function from the real numbers in the interval from 0 to 1 (including endpoints) to $\mathcal{P}(\mathbb{N})$, the power set of the natural numbers, that is total, and one-to-one, but *not* onto. (Remember that in binary notation, $0.{}_2011111\cdots = 0.{}_21$ for the same reason that $0.09999\cdots = 0.1$ in decimal notation.)

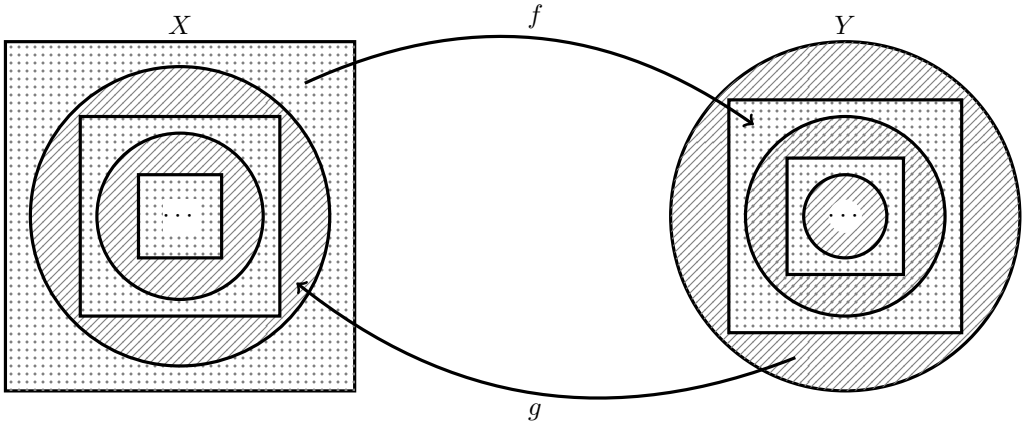
Exercise 4.7 Using notation in any base greater than 2, show that there is a function from $\mathcal{P}(\mathbb{N})$ to the set of real numbers in the interval from 0 to 1 (including endpoints), that is total, and one-to-one, but *not* onto.

These two results show that there is a correspondence from $\mathcal{P}(\mathbb{N})$ to a subset of the interval from 0 to 1, and there is also a correspondence from a subset of the interval from 0 to 1 and $\mathcal{P}(\mathbb{N})$. Intuitively, this means the interval is no larger than the power set of the naturals, and the power set of the naturals is no larger than the interval, so they should be the same size. Section 4.7 proves a theorem that shows that this reasoning is in fact valid, but there are a lot of technical details. In section 4.8 we will show that you can construct the correspondence *without* going through all those technicalities.

4.7 The Cantor-Schroeder-Bernstein Theorem

Theorem 4.2 (Cantor-Schroeder-Bernstein Theorem) If f is a function from X to Y that is total, and one-to-one, but not onto, while g is a function from Y to X that is total, and one-to-one, but not onto, then there is a correspondence between X and Y.

Proof. Consider the following diagram, illustrating the features given in the statement of the theorem. The fact that f sends X in a total, one-to-one, but not onto way into Y is illustrated by the fact that X's shape is duplicated inside Y (but not covering it), and conversely for g. Every region of a given shape and shading on one side is sent to a region of the same shape and shading on the other—but note that f sends the outermost dotted square of X to the dotted square just inside Y, while g sends that dotted square just inside Y to the *second* dotted square in X.



We will construct the correspondence between all of X and all of Y by putting these things together. We define a function h by letting it behave like f for the dotted regions of X, and letting it behave like g^{-1} for the striped regions of X. This turns out to be the correspondence.

To make this precise, and not rely on the visual aid, define subsets in the following way. Let

$$X_0 = \{x \in X \mid \neg\exists y{\in}Y(g(y) = x)\}, Y_0 = \{y \in Y \mid \neg\exists x{\in}X(f(x) = y)\}.$$

Then for every natural number n, define

$$X_{n+1} = \{x \in X \mid \exists y{\in}Y_n(g(y) = x)\}, Y_{n+1} = \{y \in Y \mid \exists x{\in}X(f(x) = y)\}.$$

Finally, define

$$X_\infty = \{x \in X \mid \neg\exists n(x \in X_n)\}, Y_\infty = \{y \in Y \mid \neg\exists n(y \in Y_n)\}.$$

Now we make some observations. Observe that if $i \neq j$, then X_i is disjoint from X_j, and X_∞ (and similarly for Y)—proving this is a tedious exercise in proof by induction. Observe also that for each i, f is a correspondence between X_i and Y_{i+1}, while g is a correspondence between Y_i and X_{i+1}—this is actually straightforward to prove (f and g are both functions that are total and one-to-one on these subsets, because they are functions that are total and one-to-one on the whole sets X and Y, and they are onto because of how X_{i+1} and Y_{i+1} are defined). Finally, note that, for the same reason, f is a correspondence between X_∞ and Y_∞, and g is a correspondence between Y_∞ and X_∞ (though they don't necessarily behave in any similar way on these sets).

At last, we construct the correspondence. Let $h(x) = \begin{cases} f(x) & \text{if } x \in X_n \text{ with } n \text{ even} \\ g^{-1}(x) & \text{if } x \in X_n \text{ with } n \text{ odd} \\ x & \text{if } x \in X_\infty. \end{cases}$

The observations we have already made ensure that h is a correspondence between X and Y.

□

4.8 Proving the correspondence *without* the Cantor-Schroeder-Bernstein Theorem

If you don't want to use the Cantor-Schroeder-Bernstein theorem to prove that there is a correspondence between $\mathbb{R}$ and $\mathcal{P}(\mathbb{N})$, we can also do it more directly. Recall that we said that a set of natural numbers is co-finite if it contains all the natural numbers except for finitely many.

In Exercise 4.6, we used binary notation to try to make a correspondence between $\mathcal{P}(\mathbb{N})$ and the set of all real numbers between 0 and 1. It turns out that if a set is neither finite nor co-finite, then the correspondence we made gives a notation that is a unique real number. But if a set is finite, it yields a notation that ends with infinitely many 0's, and if it is co-finite, it yields a notation that ends with infinitely many 1's. Note that the set of real numbers with a notation that ends with infinitely many 0's is the set of fractions whose denominator is a power of 2. And with a bit of care, we can see this must also be the same as the set of real numbers with a notation that ends with infinitely many 1's.

However, we have already shown that the set of finite sets of natural numbers is countable, and so is the set of cofinite sets, so the union of these two sets is also countable. But the set of real numbers strictly between 0 and 1 that can be written as fractions whose denominator is a power of 2 is a subset of $\mathbb{Q}$, which we have already shown is countable.

Thus, the attempted correspondence is a perfect correspondence between the set of sets of natural numbers that are neither finite nor co-finite and the set of real numbers in the interval between 0 and 1 that are not fractions with denominator a power of 2. But the sets that are left out on one side form a countable set, and the fractions that are left out on the other side also form a countable set, so there must be a perfect correspondence between those.

Putting this all together, we get a correspondence between $\mathcal{P}(\mathbb{N})$ and the set of real numbers in the interval from 0 to 1.

Now that we have shown that there is a correspondence between the set $\mathbb{R}$ of real numbers, and the power set of the natural numbers, $\mathcal{P}(\mathbb{N})$, we can return to the question, is there any other infinity that is between the size of $\mathbb{N}$ and this one? Because $\mathbb{R}$ is the "continuous" set of real numbers, Cantor called this question "the continuum problem". "The Continuum Hypothesis" is the hypothesis that there is no infinity in between these two, and its negation is the hypothesis that there is an infinity in between these two.

5 The Axiom of Choice, the paradoxes

In order to address the Continuum Hypothesis, Cantor wanted a way to more explicitly compare different infinite cardinalities, to see which is larger, and to see if there are any infinite cardinalities in between them. Cantor claimed to have made progress on this result by proving that every set has a well-ordering, which would allow earlier results about the ordinals to apply to arbitrary sets. However, this claim was quite controversial, and in 1904, the Hungarian mathematician Kőnig Gyula (publishing under the German version of his name, Julius König) claimed to have a proof that it was impossible to create a well-ordering for the real numbers. However, within a few weeks, Felix Hausdorff found an error in König's proof, and that same year, Ernst Zermelo published a clear axiomatic proof of the well-ordering theorem. However, in addition to a straightforward set of axioms that we will discuss in more detail in Part II, Zermelo had to explicitly add a controversial new principle known as the Axiom of Choice.

5.1 The Axiom of Choice

Axiom (Axiom of Choice) If X is a set of sets, and for every $S \in X$, S is non-empty, then there exists a function f such that for each $S \in X$, $f(S) \in S$.

The function f is called a "choice function" on the sets in X.

There are certain conditions under which the existence of a choice function is straightforward. For instance, if X has only one member, S, and S is non-empty, then that means there exists an element $s \in S$, and we can consider the set consisting of a single ordered pair, $\{(S, s)\}$. Based on our definition of a function as a set of ordered pairs such that there is at most one pair with any given first element, this is a function, and $f(S) \in S$. Similarly, if X has only two members, S_1 and S_2, and both are non-empty, then that means there exists $s_1 \in S_1$ and $s_2 \in S_2$, so we can consider the set consisting of two ordered pairs, $\{(S_1, s_1), (S_2, s_2)\}$, which is a choice function on this set. In general, whenever X is finite, we can just pick these members and define the set.

The difficulty arises when there are infinitely many members of X—we can't actually write down all the members then, to define the choice function.

The philosopher Bertrand Russell gave a clever illustration of a further important distinction in when the Axiom of Choice is needed. He noted that if X is an infinite set of pairs of shoes, then we can prove there exists a choice function without using the Axiom of Choice, but we need the Axiom of Choice if X is an infinite set of pairs of socks.

Russell noted that if X is a infinite set of sets, we can define the set $\bigcup X$ to be the union of all the elements of X (that is, it contains all the shoes, or all the socks, or all the things in any set in X). Then if X is a set of pairs of shoes, we can define a relation between X and $\bigcup X$ as the set $\{(S, s) \mid s \in S \text{ and } s \text{ is a left shoe}\}$. This is a choice function on X, because every set in X has exactly one left shoe in it. (We could of course also use a different choice function, $\{(S, s) \mid s \in S \text{ and } s \text{ is a right shoe}\}$.) But we can't do anything similar with the infinite set of pairs of socks—there is no property you can generally use to pick out exactly one sock from every pair.

The Axiom of Choice is only relevant when you want a choice function for an infinite set of sets, where you have no property that is guaranteed to pick out exactly one element from each set.

We can also do more. If someone tells us that f is a choice function on the infinite set of pairs of socks, we can construct infinitely many other choice functions on this set. For any particular pair S of socks, we can define f' to be the function that agrees with f on all the pairs other than S, but chooses the other sock on S. Any of these other functions could just as well have been the choice function to start. In fact, for every subset of X, we can pick a different choice function, by replacing the output of f on the elements of that set, while leaving the output the same outside that set.

So when the Axiom of Choice is needed to prove the existence of even one choice function, we can show that uncountably many such choice functions exist. The issue is like an infinite version of Buridan's Ass, the donkey that has two bales of hay in front of it, but can't choose which one to go to, and thus starves.

5.2 Zermelo's Proof

Now that we have some sort of understanding of the Axiom of Choice, we can see how Zermelo used it to prove the Well-Ordering Theorem.

Theorem 5.1 (The Well-Ordering Theorem) Every set has a well-ordering.

Proof. Let S be any set. We will construct a function g from the ordinals to S that is one-to-one and onto. This will yield a correspondence between S and some ordinal, and this correspondence defines a well-ordering of S.

To do this, start by letting X be the set of all non-empty subsets of S. By the Axiom of Choice, there is a function f that "chooses" one element of each such set. We will use this function to define g. Intuitively, we will use f to "pick" elements of S one-by-one, "until we have picked them all", and this will generate the well-ordering. Of course, this is just a metaphor—there is no actual *process* here, but thinking of it in terms of a process that runs possibly infinitely long helps understand what is going on.

Let us consider a particular set W of functions whose domain is some set of ordinals, and whose range is some subset of S. Not every such function will be in the set, but just the functions satisfying certain conditions. In particular a function h will be a member of W if it satisfies two conditions. First, for every ordinal α, if $h(\alpha)$ is defined, then so is $h(\beta)$ for all $\beta < \alpha$. Second, for every ordinal α, consider the set $S_\alpha = \{g(\beta) \mid \beta < \alpha\}$. That is, S_α is the set of all elements of S picked in the first α steps. Let $\bar{S_\alpha} = \{x \in S \mid x \notin S_\alpha\}$ be the remainder, the set of all elements of S that haven't been picked in the first α steps. The second condition on h is that for every ordinal α, if $h(\alpha)$ is defined, then $h(\alpha) = f(\bar{S_\alpha})$.

Now notice a few things. First, each $h \in W$ is one-to-one. This is because each $h(\alpha)$ is the result of using f to pick some element that hadn't occurred earlier in the function.

Second, if $h_1 \in W$ and $h_2 \in W$, then for any ordinal α for which $h_1(\alpha)$ and $h_2(\alpha)$ are both defined, it must be the case that $h_1(\alpha) = h_2(\alpha)$. (If this weren't the case, then there would have to be some *earliest* α for which these two functions differ, but then since $h_1(\beta) = h_2(\beta)$ for all $\beta < \alpha$, this would mean that the two functions would have to have the same S_α, and so both $h_1(\alpha) = h_2(\alpha) = f(\bar{S_\alpha})$.)

If we think of each function as a set of ordered pairs, then the union of all these sets of ordered pairs is also a function, h_∞ whose domain is some set of ordinals, and whose range is some subset of S. This function h_∞ must satisfy the defining property, and thus $h_\infty \in W$. That is h_∞ is the unique largest element of W.

Now let us consider the domain of h_∞. This domain must in fact be a set, since we got it from a set by just applying a function and union operations. This domain can't be the set of all ordinals, because there is no set containing all ordinals. (This was the result of the Burali-Forti paradox.)

Thus, there must be some ordinal α such that $h_\infty(\alpha)$ is undefined. Consider the set $S_\alpha = \{h_\infty(\beta) \mid \beta < \alpha\}$, as above. If $S_\alpha \neq S$, then $\bar{S_\alpha}$ is non-empty, so there would be a possibility of extending h_∞ by defining $h_{\infty+1}(\alpha) = f(\bar{S_\alpha})$. But then this $h_{\infty+1}$ would have to be in W, contradicting the definition of h_∞ as the largest element of W.

So in order for $h_\infty(\alpha)$ to be undefined we must have $S_\alpha = S$. Since h_∞ is in W, it is one-to-one, and since $S_\alpha = S$, this means h_∞ is onto. Thus, we can set $g = h_\infty$, and see that there is in fact a function from the ordinals to S that is one-to-one and onto. This defines a well-ordering of S. □

This first exercise requires you to copy part of the reasoning of Zermelo's proof:

Exercise 5.1 Show (using the Axiom of Choice) that every set either has a correspondence with the set $\{0, \ldots, n\}$ for some natural number n, or it has a subset that has a correspondence with the natural numbers.

This will allow us to tie up a loose end from earlier. We stated that there are two possible definitions of what it means to be infinite. One is just the negation of being finite, namely that there is no correspondence between the set and $\{0, \ldots, n\}$ for ay natural number n. The other is Dedekind's definition, stating that there is a correspondence between the set and some proper subset. It is straightforward to see that the natural numbers have a correspondence with a proper subset (consider the set of ordered pairs $\{(n, n+1) \mid n \in \mathbb{N}\}$, which is a correspondence between $\mathbb{N}$ and the set of non-zero natural numbers), so any set that has a subset corresponding to the natural numbers does too.

Exercise 5.2 Show (without using the Axiom of Choice, or anything complex) that if a set has a correspondence with a proper subset of itself, then there is a subset that has a correspondence with the natural numbers. (Hint: Let f be a correspondence between S and some proper subset of itself, and let 0 correspond to some element of S that is *not* in this subset.)

Thus, a set is infinite according to Dedekind's definition if, and only if, it has a subset with a correspondence to the set of natural numbers. So the Axiom of Choice proves that the two definitions of infinity are equivalent.

5.3 Applications of the Well-Ordering Theorem

Exercise 5.3 Show (using the Well-Ordering Theorem, and some facts about ordinals from before) that for every set, there is a *least* ordinal that it has a correspondence with.

Exercise 5.4 Use the previous result to show that for every two sets, one of them has a correspondence with a subset of the other (this might not be a proper subset, if the sets have the same cardinality).

Cantor used the Hebrew letter Aleph, "ℵ", to stand for these ordinals. (Most of the Greek and Roman letters were already used for other concepts in mathematics, including Cantor's use of ω, and since Cantor was Jewish, he was familiar with the Hebrew letters.) He let $\aleph_0$ be the smallest infinite cardinal number (which is equal to ω), and then let $\aleph_1$ be the next cardinal number, and then $\aleph_2$ the next one, and so on. Because each of these is well-defined, there must be a set $\{\aleph_n \mid n \in \mathbb{N}\}$, and the union of this set would have to be an ordinal larger than any of them, which must be $\aleph_\omega$. But this is not the largest infinite cardinality (since the power set of any set has a larger cardinality than that set), so the next one must be $\aleph_{\omega+1}$, and then $\aleph_{\omega+2}$, and so on. A similar argument shows that in fact, for every ordinal α, there is a cardinal $\aleph_\alpha$.

Returning to Cantor's continuum problem, we see that the Continuum Hypothesis says that the cardinality of $\mathbb{R}$ is $\aleph_1$. But if the Continuum Hypothesis is false, the cardinality of $\mathbb{R}$ could be $\aleph_2$ or $\aleph_3$ or $\aleph_{17}$ or $\aleph_{\omega+3}$. In fact, Paul Cohen proved in 1963 that modern axiomatic set theory is not enough to answer the continuum problem. In fact, in 1970s, Easton's Theorem showed that the size of the continuum could be nearly any ℵ. It can't be precisely $\aleph_\omega$, or $\aleph_{\omega+\omega}$, or anything else like that (in a sense that can be made precise), but it can be anything else.

5.4 Finite games with hats

Let's take a break from infinite sets for a bit.

Let's consider a game, where n people stand in a circle. I will either put a black hat or a white hat on every person's head. Everyone can see the hat on everyone else's head, but not the hat on their own head. Now the players will go around the circle, and take turns guessing the color of the hat on their own head.

If everyone is just guessing at random, you can see that on average, about half of the people are going to guess incorrectly. But it turns out that if they plan ahead, they can do better. In fact, there's a strategy they can take that will guarantee that at most one person will get their guess wrong!

The important thing to think about here is that not only can everyone see everyone else's hat, they can also *hear the guesses* of everyone who guesses before them. So early people can use their guesses to communicate information to later people.

The basic idea is that the first person should look out at all the hats they can see, and should guess "white" if they see an odd number of white hats, and "black" if they see an even number of white hats. (The reverse would also work, but the players need to have agreed in advance which of these strategies they will use.) Once the first player has made their "guess", everyone else can look around at what hats they can see. They ignore the first person's hat, since they know that person can't see their own hat, but counting the rest, they can either see an odd number of white hats or an even number of white hats. If the first person saw an odd number of white hats, and this person sees an odd number of white hats on heads other than the first person's, then they know their own hat must be black. But if the first person saw an odd number of white hats, and this person sees an even number of white hats on heads other than the first person's, then

they know their own hat must be white. (Similar reasoning works if the first person saw an even number of white hats.)

Thus, if they agree on one of these strategies, then as soon as the first person makes their "guess", everyone else can look around and know the color of their own hat. Thus, everyone but the first player can guess their own hat color correctly—and even the first player will sometimes get their own color right just by chance.

Exercise 5.5 Show that for any n, these (i.e., first player guesses white if they see an odd number of white hats; first player guesses black if they see an odd number of white hats) are the *only* two strategies that can guarantee that everyone but the first player guesses correctly.

However, if they all have to guess at once (using just the information about hats other than their own), we can prove that no matter what strategy they use, there must be some possibilities where at least half of them get their guess incorrect. If there are any possibilities where a majority of them get their guess correct, then there must also be possibilities where a majority of them get their guess incorrect.

To see this, we will count the possible number of correct or incorrect guesses in two different ways. There are 2^n ways of assigning hat colors to the players, and each of the n players makes a guess on each such assignment, so there are $n \cdot 2^n$ total possible guesses. For a given player, we can see that regardless of the strategy, they must get their own hat right in exactly half of the different possible assignments. That's because, if we keep everyone else's hat fixed, we must keep a player's guess fixed, and there is exactly one possibility where that results in a correct guess for the player, and exactly one possibility where that results in an incorrect guess for the player. Thus, the total number of correct guesses across all 2^n possibilities must be precisely $\frac{n \cdot 2^n}{2}$.

If it were true that, in every possibility, a majority of players got their guess correct, then there would have to be more than $n/2$ correct guesses in each of the 2^n possibilities. Thus, there must be some possibilities where at least half of players get their guess incorrect. Furthermore, if there are *any* possibilities where more than $n/2$ guesses are correct, there must be some other possibilities where *less* than $n/2$ guesses are correct, in order that the total can be precisely $\frac{n \cdot 2^n}{2}$. Depending on their strategy, the players can affect how they cluster these correct and incorrect guesses—they can ensure that *usually* a majority will get it right, but then sometimes they will all get it wrong, and they will never all be right.

5.5 Infinite games with hats

But look what happens if there are infinitely many players. Assume that there are $\aleph_0$ many players, i.e., one player for each natural number, and that each can see the hat on everyone's head but their own, and they all have to guess at the same time. Using the Axiom of Choice, we can prove that there is a strategy that guarantees that *always*, not only a *majority* of players get their guess right, but in fact *only finitely many* get it wrong!

The way they do this is the following. Consider the set of all possible assignments of hats to the players. Say that two of these assignments are "similar" if there are only finitely many differences between the assignments. For any assignment x, let S_x be the set of all assignments similar to x. We can see that if x is similar to y, then $S_x = S_y$ (anything similar to x is similar to y), while if x is not similar to y, then $S_x \cap S_y = \emptyset$ (nothing is both similar to x and similar to y). Call these sets the "similarity classes".

Now consider the set X of all similarity classes of assignments. Each of these similarity classes is non-empty, so the Axiom of Choice tells us there is a function f such that $f(S) \in S$ for any $S \in X$.

Here is the strategy that each player will use. When a player looks out at all the other hats, they can't tell precisely which assignment x is actual. However, they can tell which similarity class S_x the actual assignment is in. Then $f(S_x)$ is some assignment within this similarity class. This assignment gives a color to each player. Each player then guesses that their hat matches this assignment. Although it is unlikely that every player guesses their own hat correctly, we can see that the true assignment must be in the same similarity class as the guess (because every player can see what similarity class they are in, even if they can't see the total true assignment). Which means that the actual hat colors must only differ from the guessed hat colors for finitely many players. That is, all but finitely many of them get it right.

What's even more amazing about this is that it doesn't matter how many possible hat colors there are! It works just as well if there are 2 colors, or 17 colors, or as many colors as natural numbers, or even some gigantic uncountable infinity of possible colors, most of which aren't assigned to anyone in any given run of the game! No matter how big the set of possible colors is, there is a set of possible assignments, and a set of similarity classes, and each player can see which similarity class they are in.

It's because of results like this that the Axiom of Choice is controversial.

The most famous such example is the Banach-Tarski "Paradox". I call it a "Paradox", because it is actually a theorem, and doesn't lead to any contradiction—it is just another disturbing result like the one about the game with hats. The full proof of the theorem requires a bunch of mathematics (topology, geometry, group theory) that is beyond the scope of this class. But I can state it here.

Theorem 5.2 (Banach-Tarski "Paradox") Let S be the set of points in a sphere. There exist S_1, S_2, S_3, S_4 that are disjoint subsets of S whose union is S, with the following property: there exist rotations ρ_1 and ρ_2 such that $\rho_1(S_1) \cup \rho_2(S_2) = S$, and there exist rotations ρ_3 and ρ_4 such that $\rho_3(S_3) \cup \rho_4(S_4) = S$.

That is, a sphere can be "cut" into finitely many pieces, and the pieces can be put back together into two full copies of the sphere, with no points missing. Of course I say "cut", because these pieces are strange and diffuse sets of points scattered throughout the entire sphere, just like the "strategy" in the infinite game with hats.

But to distinguish these strange results that have the property of being logically consistent, and true if the Axiom of Choice is, from the paradoxes we discussed in earlier sections, we will now start to introduce the rest of Zermelo's axioms and show how they give rise to the results we have used so far, but avoid the paradoxes.

Part II: Formal axiomatic set theory

6 The formal language, and the "small" axioms: Extensionality, Pairing, Union, and Empty Set

In order to avoid the paradoxes, and ensure that we know precisely what is provable, we will make every assumption about set existence explicit.

Along the way, we will need to talk about a formal language that shows what we can express precisely in the language of set theory. This will be made more explicit as we get to that, but for those of you familiar with first-order logic, this formal language will include the relations "$\in$" and "$=$", and everything definable from these using the logical connectives "and" ($\wedge$), "or" ($\vee$), "not" ($\neg$), and the quantifiers "there exist" ($\exists$) and "for all" ($\forall$). We can also use the logical connective "if φ then ψ", defined as "$\neg(\varphi \wedge \neg\psi)$" or "$(\neg\varphi \vee \psi)$".

We have the intuitive idea that "$\in$" means "is a member of", but this is only the guiding intuition. We will try to make the meaning more precise by stating *axioms*, which are the basic facts that tell us everything we can say formally about sets and membership. But because the language with just "*in*" and "$=$" is somewhat restricted, we will also supplement this language with many *definitions*—each definition introduces a new symbol, and tells us everything there is to know about when that symbol applies. Anything we express using the new symbol could be expressed without it

Definition "$x \subseteq y$" is an abbreviation for "every element of x is an element of y" (or, in the formal language, $\forall z(z \in x \to z \in y)$).

Exercise 6.1 If $x \subseteq y$ and $y \subseteq z$ then $x \subseteq z$.

Axiom (Pure Set Extensionality) $x = y$ if x and y have the same elements.

Within the formal language, we could state this as $x = y \leftrightarrow \forall z(z \in x \leftrightarrow z \in y)$, and if we make use of the definition of "$\subseteq$", we could even state it as $x = y \leftrightarrow (x \subseteq y \& y \subseteq x)$. This axiom is just another way of stating something that we already stated back in the beginning, that two sets are actually the same set if they have the same elements, but now we are making explicit that this is one of the axioms of Zermelo-Fraenkel set theory, that tells us some of the basic facts about what sets are.

Notice that, among other things, it implies that there can be at most one object with no elements. (Any two objects that both have no elements would have the same elements, and would thus have to be the same object.) This means that, in this context, everything we are talking about is a set—the members of sets are themselves sets.

Axiom (Ur-Element Extensionality) If x and y are sets, then $x = y$ if they have the same elements.

Axiom (Set of Ur-Elements) There is a set that contains everything that is not a set.

There is an axiom that states that there is in fact an object that has no elements. This axiom turns out to be implied by axioms we will discuss later, so it is sometimes left out of some versions of the Zermelo-Fraenkel axioms, but it is useful to include it.

Axiom (Empty Set) There is a set that has no elements.

Definition "$\emptyset$" is a name for the set that has no elements.

Axiom (Pairing) For any x and y, there is a set that contains x, and contains y, such that any element of that set is either x or y.

Definition "$\{x, y\}$" is a name for the set that contains x, and contains y, such that any element of that set is either x or y.

Notice that this axiom (and the associated definition) doesn't require that x and y be distinct objects. Thus, among the implications of this axiom is the claim that for every object x, there is a set whose *only* element is x. Such a set is called a *singleton*.

Definition "$\{x\}$" is a name for the set that contains x, and nothing else.

It is straightforward to use these two axioms (and the relevant definitions) to see that $\{x, y\} = \{y, x\}$.

If x and y exist, then the above axioms and definitions let us conclude that $\{x\}$ and $\{x, y\}$ exist, and so the Pairing axiom lets us conclude that $\{\{x\}, \{x, y\}\}$ exists as well.

Definition (Kuratowski's definition) "$\langle x, y\rangle$" is an abbreviation for "$\{\{x\}, \{x, y\}\}$".

This is how we make precise the concept of an ordered pair, which we just took for granted before. One thing to note, which is slightly strange, is that if $x = y$, then $\langle x, y\rangle = \{\{x\}\}$.

Exercise 6.2 $\langle x, y\rangle = \langle z, w\rangle$ iff $x = z$ and $y = w$.

Axiom (Union) For any X, there is a set that contains all and only the elements of any sets that are elements of X.

Definition "$\bigcup X$" is a name for the set that contains all and only the elements of any sets that are elements of X.

Definition "$x \cup y$" is an abbreviation for "$\bigcup\{x, y\}$".

The relation between the symbols "$\bigcup$" and "$\cup$" is a lot like the relation between the mathematical symbols "$\sum$" and "$+$".

Notice that if $x \in X$ then $x \subseteq \bigcup X$.

Using the Pairing axiom, we can show that for any x, y, z, there are sets $\{x, y\}$ and $\{z\}$, and thus $\{\{x, y\}, \{z\}\}$. Then by the Union axiom, there is a set $\bigcup\{\{x, y\}, \{z\}\}$, and we can see that the elements of this set are all and only x, y, z. We can similarly do this for any other finite number of objects.

Definition For any finite list of objects $x_1, x_2, \ldots, x_n$, "$\{x_1, x_2, \ldots, x_n\}$" is a name for the set that contains these objects as members, and nothing else.

Thus, although Pairing only directly puts two elements together into a set, we can use Pairing and Union together to create sets with any specific finite set of elements.

Definition "$S(x)$" is an abbreviation for "$x \cup \{x\}$".

This lets us generate a particular sequence of sets that we will use to stand for the natural numbers. In particular, we will use the numeral "0" to stand for $\emptyset$, and for any natural number n, we will use the numeral $n + 1$ to stand for $S(n)$.

In order for this definition to make sense, we need to make sure that every set that arises in this sequence is different from all the earlier sets.

First, observe that if x has no element that is an element of itself, then x is not an element of itself. (After all, if x were an element of itself, then it would have an element that is an element of itself, namely x!)

Now note that if x has no element that is an element of itself, then $S(x) \neq x$, because x is an element of $S(x)$, but is not an element of x. Furthermore, $S(x)$ is not equal to any element of x, because it contains every element of x, and we assumed that no element of x contained itself. Finally, $S(x)$ also has no element that is an element of itself, because its elements are just the elements of x, and x itself, none of which is a member of itself.

Now, it's clear that 0 has no element that is an element of itself (because it has no elements at all!) so the above results apply. Thus $S(0) = 1$ is different from 0, and has no element that is an element of itself. Thus $S(1) = 2$ is different from 1, and from every element of 1, and has no element that is an element of itself. And similarly for $3, 4, 5, \ldots$. Each new natural number is different from each previous natural number, and has no element that is an element of itself.

Exercise 6.3 Using these definitions, show that each n is the set of all natural numbers strictly less than n. (This will be a proof by induction. The base case is to show that this is true when $n = 0$, where there is no natural number less than n, and the induction step involves showing that if it's true for n, then it's also true for $n + 1$.)

Note that this shows that each natural number n has exactly n members. These results also mean that there are infinitely many distinct sets—though we can't yet prove that there is a set that itself has infinitely many distinct elements.

7 The "medium" axioms: Power set, (Foundation), and Separation

7.1 Power set

Axiom (Power Set) For any set x, there is a set whose elements are all and only the subsets of x.

Definition "$\mathcal{P}(x)$" is a name for the set whose elements are all and only the subsets of x.

Exercise 7.1 Find $\mathcal{P}(\emptyset)$, $\mathcal{P}(\mathcal{P}(\emptyset))$, and $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$ and write them explicitly using curly braces.

Note that for any x, both $\emptyset \in \mathcal{P}(x)$ and $x \in \mathcal{P}(x)$, but you should have already shown in the above exercise that not every power set has at least two elements. (Can you identify all the sets whose power set has fewer than two elements?)

Exercise 7.2 If $x \subseteq y$ then $\mathcal{P}(x) \subseteq \mathcal{P}(y)$.

Exercise 7.3 $x = \bigcup \mathcal{P}(x)$ and $x \subseteq \mathcal{P}(\bigcup x)$, but there exists a set x such that $x \neq \mathcal{P}(\bigcup x)$.

We will eventually be able to prove Cantor's result that the power set of a set is always bigger than the set, but in the axiomatic context, it will take some time to set up the precise formal statement of what that means.

7.2 Foundation

The next axiom is usually included for convenience, but isn't essential to the main mathematical results. Other than Extensionality, it is the only axiom that doesn't take the form of constructing a new set out of an old set.

Axiom (Foundation) For any non-empty set x, there is a member of x that has no elements in common with x.

This axiom is somewhat strange, but the most important implications of this axiom can be expressed more directly.

First, this axiom ensures that there is no set S with $S \in S$. If there were, we could then consider the set $x = \{S\}$. This set has only one element, namely S. But S has an element in common with x, namely S itself!

Second, this axiom ensures that there is no finite sequence $S_0 \in S_1 \in \cdots \in S_n \in S_0$. Again, if there were, then our earlier results with Pairing and Union would let us construct a set $x = \{S_0, S_1, \ldots, S_n\}$. But every element of this set has at least one element in common with this set, namely the earlier element.

Finally, this axiom also entails that there is no set of all sets—if there were such a set, it would contain itself. But we will use other means to show that there is no set of all sets.

7.3 Separation

Axiom (Separation) For any set x, and any formula $\phi(w)$ in the formal language, there is a set consisting of all and only the elements of x that satisfy the formula $\phi(w)$.

Definition "$\{w \in x \mid \phi(w)\}$" is a name for this set.

This axiom justifies *some* uses of the "set-builder" notation we defined early on.

Notice that since there is a set of all ur-elements, we can use this axiom to define a set of all ur-elements but one, or all ur-elements but two.

However, the axiom of Separation doesn't let us define a set of *all* objects satisfying a property, just the objects satisfying this property that come from a set that already exists. This restriction is one of the

main ways that the Zermelo-Fraenkel axioms avoid most of the paradoxes. Several of the paradoxes involved collecting together *all* objects of a given sort, like the Burali-Forti Paradox and Russell's Paradox.

The fact that $\phi(w)$ has to be expressible in the formal language solves the other paradoxes. The Sorites paradox involved vague language that isn't part of the language of set theory, and many of the paradoxes involved the concept of "definability" or "meaning", which are not themselves definable within the formal language, like Grelling's paradox (the word "heterologous"), and Richard's paradox (the definable diagonal of the definable sets of natural numbers).

Separation lets us construct various other sets. For instance, we can construct $x \setminus y$, the set of all elements of x that are not in y, and we can construct $x \cap y$, the intersection of x and y. In fact, we can construct $\bigcap X$, the intersection of all the sets in X, and even $\bigcap \varphi(w)$, the intersection of all sets that satisfy $\varphi(w)$! That's because being in every set satisfying $\varphi(w)$ is a property that can be expressed in the formal language, and if we start with any one set satisfying this property, it will contain all the elements that are needed.

Exercise 7.4 Use the Separation axiom to prove that the empty set exists, using only the assumption that at least one set exists.

Exercise 7.5 Use the Separation axiom to prove that there is no set of all sets. (Hint: Use Russell's paradox, letting $\varphi(w)$ be "$\neg(w \in w)$".)

7.4 Developing Cantor's theory of sizes of infinity

Once we have this we can develop all the machinery for Cantor's theory of infinity, and be pretty confident we are safe from the paradoxes.

We need to say what it is for sets to have the same size - this means we need to be able to talk about the existence of correspondences between the sets, so we have to think of correspondences as sets of a particular sort.

Correspondences are relations that satisfy four properties, so we need to be able to think of relations as sets of a given sort, describe these properties in the formal language, and show that the relevant ones exist.

Relations are going to be thought of as sets of ordered pairs. We have already shown that every particular ordered pair exists, so what we will do in the rest of this section is show that the relevant sets of ordered pairs exist, and then show that we can talk about these properties, and eventually prove Cantor's theorem that the power set of a set is larger than the set. We don't yet have the existence of an infinite set, but once we add that, we will have the whole theory.

We can also use Separation to construct lots of other interesting sets. The set of all singletons from X can be constructed from the power set of X, using the formula $\varphi(w)$ given by "$\exists x \forall y(y \in w \leftrightarrow y = x)$", which says there is an object x such that x is an element of w and all elements of w are x. The set of all ordered pairs of elements of X can be constructed from the power set of the power set of X, using the formula $\varphi(w)$ stating that w is an ordered pair of two elements of X (it's worth checking that this can be expressed in the formal language). The set of all ordered pairs of an element of X with an element of Y, which we write as $X \times Y$, the *cartesian product* of X and Y, can be constructed from the power set of the power set of the union of X and Y.

Any subset of this cartesian product is called a binary relation from X to Y. For any $R \subseteq X \times Y$, we sometimes write xRy for $\langle x, y \rangle \in R$.

Again using Separation, for any X, we can construct the identity relation on X, $\{\langle x, x \rangle \mid x \in X\}$.

For any set of ordered pairs, we can use Union follows by Separation to construct the domain and range of this set, so that every set of ordered pairs can be thought of as a relation.

Putting the cartesian product in reverse, we can then use Separation to create the inverse of a relation.

If R is a relation from X to Y and S is a relation from Y to Z, then we can generate the composition $R \circ S$ as a relation from X to Z.

Say that R is a *function* if for every $x \in X$, if $\langle x, y_1 \rangle \in R$ and $\langle x, y_2 \rangle \in R$, then $y_1 = y_2$.

Say that R is *total* if for every $x \in X$, there exists a y with $\langle x, y \rangle \in R$.

Say that R is *one-to-one* if for every $y \in Y$, if $\langle x_1, y \rangle \in R$ and $\langle x_2, y \rangle \in R$, then $x_1 = x_2$.

Say that R is *onto* if for every $y \in Y$, there exists an x with $\langle x, y\rangle \in R$.
If $R \subseteq X \times Y$ and $S \subseteq Y \times Z$, then we define $R \circ S \subseteq X \times Z$ by $R \circ S = \{\langle x, z\rangle \mid \exists y(\langle x, y\rangle \in R \& \langle y, z\rangle \in S)\}$.

Exercise 7.6 If R is a relation from X to Y, and S is a relation from Y to Z, and R and S are both (functions/total/one-to-one/onto) then so is $R \circ S$ as a relation from X to Z.

By taking $\mathcal{P}(X \times Y)$ and using Separation, we can construct the set ${}^{X}Y$ of all total functions from X to Y.

Exercise 7.7 (Cantor's Theorem) Show that for any set X, there is no relation with domain X and range $\mathcal{P}(X)$ that is both a function and onto.

In the proof of Cantor's Theorem, you will need to be explicit about how, for any relation with domain X and range $\mathcal{P}(X)$ that is a function, you can use Separation to define a diagonal set, which is not part of the relation.

8 Infinity, formalized ordinals, and the well-ordering principle

Recall that we defined the successor function by $S(x) = x \cup \{x\}$. We say that a set z is *inductive* if $\emptyset \in z$ and $\forall x(x \in z \to S(x) \in z)$. That is, a set is inductive if it contains the empty set and contains the successor of anything it contains.

Axiom (Infinity) There is an inductive set.

Recall that the Separation axiom allows us to take the intersection of any definable collection of sets that has at least one member. Thus, the Axiom of Infinity allows the following definitions:

Definition ω is the intersection of all inductive sets.

Definition For x to be a *natural number* is for it to be a member of ω.

Notice that these definitions ensure that proof by induction works to establish that all natural numbers have a given property. If $\varphi(w)$ is some property, Separation allows us to consider the set of natural numbers that satisfy that property, $\{w \in \omega \mid \varphi(w)\}$. Call this set X. It is clear that $X \subseteq \omega$. But if we can show that $\varphi(\emptyset)$ and $\forall w(\varphi(w) \to \varphi(S(w)))$, then we can see that X must in fact be an inductive set. But because ω is defined to be the intersection of all inductive sets, this means that $\omega \subseteq X$. And thus, by Extensionality, since ω and X are both subsets of each other, they must be the same set. So we can conclude that every natural number has property $\varphi(w)$.

When we first discussed proof by induction, we just noted it as a method that intuitively should work. But now we have defined the natural numbers in a way that ensures that this method works, by definition.

Exercise 8.1 No natural number has a correspondence to a proper subset of itself. (Hint: The best way to prove this in the formal axiomatic context is to show that the set of natural numbers that don't have a correspondence to a proper subset is inductive.)

Exercise 8.2 ω does have a correspondence to a proper subset of itself.

Definition For x to be *finite* is for it to have a correspondence to some natural number.

Because no natural number has a correspondence with a subset of itself, it's clear that there is no more than one natural number with a correspondence to any given set.

8.1 Addition of natural numbers

We can define addition, multiplication, and exponentiation on the natural numbers in two different ways. Both ways are useful, and we can prove that they yield the same definition. Both methods show that these relations are definable in the formal language of set theory, but they have different advantages.

One method is *combinatorial*—we show some operation on sets that corresponds to the intuitive meaning of the operation, and identify the natural number that has a correspondence to the resulting set. The other method is *recursive*—we define how the operation works for the number 0, and then define how the operation works on $S(n)$ in terms of how it works on $S(n)$.

For addition, the combinatorial definition involves the union of disjoint sets: when you visualize what "13+15" means, you probably visualize something like a set of 13 students, and a separate set of 15 students being combined. It's important that these sets are disjoint—if there's one class with 13 students, and another class with 15 students, there may be less than 28 students when you combine them, if there are some students who are in both classes. To define the sum of m and n, we need to define two disjoint sets of these sizes. In order to do this, we will take advantage of the fact that ordered pairs with different second elements are always different. We will combinatorially define $m + n$ as the natural number that has a correspondence with $m \times \{0\} \cup n \times \{1\}$.

For addition, the recursive definition is as follows: we define $m + 0 = m$, and we define $m + S(n) = S(m + n)$. As I will illustrate more later, we can think of this sort of definition as saying that to add n to

m, you start with m (because $m + 0 = m$) and then apply the successor operation n times (because each time we add one more S to n, we add one more S to the output).

The first thing to note about the recursive definition is that it has two parts—there is one part that tells us how to add 0 to any number, and another part that tells us how to add the successor of n to any number, if we already know how to add n to that number. The second part of that definition might *feel* circular, but it isn't—it shows how addition of a bigger number can be defined in terms of addition of a smaller number, and if we follow these definitions back, we can eventually get down to the explicit definition with 0. It is straightforward to prove by induction that this recursive definition gives a unique answer for any $m + n$—if $n = 0$, then obviously only the first clause applies so the definition is unique; and if $m+n$ is defined uniquely, then for $m + S(n)$, only the second clause applies, and the answer will be defined, and unique.[2]

Once we have given both the combinatorial and the recursive definition of addition, we can prove that they do in fact define the same function. To do this, we work by induction. We first prove the base case, that the combinatorial definition of $m + 0$ yields the value m. (Think about what the combinatorial definition of $m + 0$ is.) And then we prove the induction step, that the number that is the combinatorial definition of $m + S(n)$ is the successor of the number that is the combinatorial definition of $m + n$.

One feature of the recursive definition is that we can use it to algorithmically reduce any expression involving 0, the successor operation, and addition, to one just involving 0 and the successor operation. For instance:

$$\begin{aligned} S(S(0)) + S(S(S(0))) &= S\Big(S(S(0)) + S(S(0))\Big) \\ &= S(S\Big(S(S(0)) + S(0)\Big)) \\ &= S(S(S\Big(S(S(0)) + 0\Big))) \\ &= S(S(S(S(S(0))))) \end{aligned}$$

The first three steps in that calculation use the fact that the term on the right side of the addition operation begins with a S (which can then be moved outside the addition), and the last step uses the fact that an addition involving 0 is trivial. The same can be done when there are multiple addition operations in the expression.

Exercise 8.3 Use a similar sequence of operations to show that $S(S(0)) + (S(0) + S(0)) = S(S(S(S(0))))$ (that is, that $2 + (1 + 1) = 4$).

We can naturally think of the combinatorial definition as giving the "meaning" of addition, its semantic content, while the recursive definition gives a syntactic process for calculating the outcome. It's useful to compare this to the way that we all learned to add numbers using Arabic numerals—think about how you figure out that $176 + 247 = 423$ ($6 + 7 = 13$, write down the 3, carry the 1, $1 + 7 + 4 = 12$, write down the 2, carry the 1, $1 + 1 + 2 = 4$, write that down, see 423). You don't have to know what these numbers mean, or that addition corresponds to taking two disjoint sets of objects and finding their union, and you can even arrange for a small set of electrical circuits, or even mechanical switches, to do this automatically without any understanding, as in the earliest mechanical calculators built in the 1700s, and widely available as cash

[2]This footnote outlines the proof of the general theorem for recursive definition of functions of natural numbers. A similar proof allows us to prove this for arbitrary functions of ordinals.

We can in fact show a general theorem about recursive definitions of this sort. Consider the set of all ordered triples of natural numbers, $(\omega \times \omega) \times \omega$. Take the set of all subsets of this set. Use Separation to remove all the sets that contain a triple whose second member is 0 and whose third member is non-zero. Use Separation again to remove all the sets that contain a triple of the form $(m, S(n), \ell)$ that don't also contain a triple of the form (m, n, k), where $\ell = S(k)$. We can prove that the remaining sets of triples will all define partial functions that satisfy the recursive definition of addition. We can prove, by induction, that for any two such partial functions, if one of them contains a triple of the form (m, n, k) and the other contains a triple of the form (m, n, ℓ), then $\ell = k$. The union of all these sets will then define a function that satisfies these conditions as well. Finally, we prove by induction that for every m, n, this union has a triple of the form (m, n, k).

This general proof strategy works for a recursive definition of *any* function $f(m, n)$, if there is some one-place function g that we define $f(m, 0) = g(m)$, and some three-place function h that we define $f(m, S(n)) = h(m, n, f(m, n))$. We will use this for multiplication and exponentiation as well.

registers in shops by the 1890s. It's a lot more complicated to write out the rules for doing addition with Arabic numerals than to write out the two simple rules that $m + 0 = m$ and $m + S(n) = S(m + n)$—but it's much easier for humans to tell the difference between "6" and "7" than between "$S(S(S(S(S(S(0))))))$" and "$S(S(S(S(S(S(S(0)))))))$".

The combinatorial definition of addition allows us to straightforwardly prove some general properties that addition has. For each of the following exercises, you should recall that "$m + n$" is defined as the natural number that has a correspondence with $m \times \{0\} \cup n \times \{1\}$, and show that there is a correspondence between the relevant sets in these exercises, which has the consequence that whatever natural number has a correspondence with one must have a correspondence with the other.

Exercise 8.4 (Commutativity of addition) For all natural numbers m, n, $m + n = n + m$.

Exercise 8.5 (Associativity of addition) For all natural numbers ℓ, m, n, $\ell + (m + n) = (\ell + m) + n$.

We can also prove these facts using the recursive definition, but it turns out to be much harder, and needs to be done by a double (or possibly triple) induction. Don't worry about following this proof (you can skip straight ahead to the section on multiplication instead), but just notice how much longer and less pleasant this is. It's much nicer to prove that the recursive definition agrees with the combinatorial definition, and then prove that the combinatorial definition satisfies commutativity.

Theorem 8.1 (Commutativity of addition, based on the recursive definition) For all natural numbers m,, $m + n = n + m$.

Proof. This will be a proof by induction on n. We need to show a base case (that 0 commutes with all numbers, i.e., for all m, $m + 0 = 0 + m$) and an inductive step (that if n commutes with all numbers, then $S(n)$ does as well).

Base case: We need to show that 0 commutes with all natural numbers, i.e., for all natural numbers m, $m + 0 = 0 + m$.

> *Proof.* The recursive definition of addition tells us that $m+0 = m$. So to prove this base case, we need to show that $0+m = m$. You might have hoped that this directly follows from the recursive definition, but unfortunately the recursive definition only tells us what to do with numbers on the *right* side of the "+" sign, not the left. So to prove that $0 + m = m$, we need to work by induction, this time on m.
>
> > Base case: We need to show that $0 + m = m$ when $m = 0$, i.e., we need to show that $0 + 0 = 0$. This does follow directly from the recursive definition of addition.
> >
> > Inductive step: Now we need to show that if $0 + m = m$, then $0 + S(m) = S(m)$. By the recursive definition of addition, $0 + S(m) = S(0 + m)$. So if $0 + m = m$, then we can replace this inside the parentheses, so that $0 + S(m) = S(m)$.
>
> Thus, we have shown that for all natural numbers m, $0 + m = m$. Since $m + 0 = m$, we now know that for all m, $m + 0 = 0 + m$, so 0 commutes with all natural numbers. □

Inductive step: We need to show that if n commutes with all numbers, then $S(n)$ does as well.

> *Proof.* We want to show that $S(n)$ commutes with all numbers, i.e., that for all m, $m + S(n) = S(n) + m$. We will do this by yet another induction, this time on m. That is, we need to show a base case (that if n commutes with all numbers, then 0 commutes with $S(n)$) and an inductive step (that if n commutes with all numbers, and m commutes with $S(n)$, then $S(m)$ commutes with $S(n)$).
>
> The base case just requires us to show that 0 commutes with $S(n)$. Fortunately, above we showed that 0 commutes with all numbers, so we don't need to prove this again.
>
> For the inductive step, we need to show that if n commutes with all numbers, and m commutes with $S(n)$, then $S(m) + S(n) = S(n) + S(m)$.

By the recursive definition of addition, we know that $S(m) + S(n) = S\Big(S(m) + n\Big)$. By the assumption that n commutes with all numbers, this equals $S\Big(n + S(m)\Big)$, and by the recursive definition of addition, this equals $S(S\Big(n+m\Big))$. Because n commutes with all numbers, $S(S(n+m)) = S(S(m+n))$, and by the recursive definition of addition, this equals $S(m+S(n))$. By the assumption that m commutes with $S(n)$, this equals $S(S(n)+m)$, and by the recursive definition of addition, this equals $S(n) + S(m)$, and we are done with this inductive step.

Thus, we have shown that if n commutes with all numbers, then so does $S(n)$. □

Thus, we have shown that 0 commutes with all numbers, and that if n commutes with all numbers, then $S(n)$ commutes with all numbers. So this completes the big induction, allowing us to conclude that all numbers commute. □

8.2 Multiplication of natural numbers

For multiplication, the combinatorial definition involves the cartesian product: the set of ordered pairs. When you visualize what "$4 \cdot 6$" means, you probably visualize something like the number of squares in a rectangular grid with 4 rows and 6 columns. Each row can be numbered with a natural number below 4, and each column can be numbered with a natural number below 6, and each square can be labeled with the corresponding ordered pair.

$\langle 0,0\rangle$	$\langle 0,1\rangle$	$\langle 0,2\rangle$	$\langle 0,3\rangle$	$\langle 0,4\rangle$	$\langle 0,5\rangle$
$\langle 1,0\rangle$	$\langle 1,1\rangle$	$\langle 1,2\rangle$	$\langle 1,3\rangle$	$\langle 1,4\rangle$	$\langle 1,5\rangle$
$\langle 2,0\rangle$	$\langle 2,1\rangle$	$\langle 2,2\rangle$	$\langle 2,3\rangle$	$\langle 2,4\rangle$	$\langle 2,5\rangle$
$\langle 3,0\rangle$	$\langle 3,1\rangle$	$\langle 3,2\rangle$	$\langle 3,3\rangle$	$\langle 3,4\rangle$	$\langle 3,5\rangle$

Since m is the set of natural numbers below m, and n is the set of natural numbers below n, we combinatorially define $m \cdot n$ as the natural number that has a correspondence with $m \times n$, the set of ordered pairs from m and n.

One convenient fact to notice—the set used to define $m + m$ is the same as the set used to define $m \cdot 2$. Both of them include every ordered pair whose first element is a member of m, and whose second element is either 0 or 1.

For multiplication, the recursive definition works as follow. We define $m \cdot 0 = 0$, and we define $m \cdot S(n) = m \cdot n + m$. The same argument as above shows that this definition is sufficient to give a unique answer. We can also see that this definition basically tells us that multiplying m by n means starting with 0, and then, n times, applying the operation of adding m.

Exercise 8.6 Prove (by induction on n) that the combinatorial definition of multiplication and the recursive definition agree for all m, n. (The base case will be very straightforward; for the induction step, there is a combinatorial definition of $m \cdot S(n)$, and there is a combinatorial definition of $m \cdot n + m$, and you will want to demonstrate a correspondence between the relevant sets in this combinatorial definition.)

Again, we can use this recursive definition to first eliminate each S on the right side of a multiplication and replace it by additions, and then replace a multiplication by 0 with a 0, and then pull out each S on

the right side of an addition, and cancel 0's on the right side of addition:

$$\begin{aligned}
S(S(0))) \cdot S(S(0)) &= \Big(S(S(0))) \cdot S(0)\Big) + S(S(0))) \\
&= \Big(\big(S(S(0))) \cdot 0\big) + S(S(0)))\Big) + S(S(0))) \\
&= \Big(0 + S(S(0))\Big) + S(S(0)) \\
&= S\Big(0 + S(0)\Big) + S(S(0)) \\
&= S(S\Big(0 + 0\Big) + S(S(0)) \\
&= S(S(0)) + S(S(0)) \\
&= S\Big(S(S(0) + S(0)\Big) \\
&= S(S\Big(S(S(0)) + 0\Big)) \\
&= S(S(S(S(0))))
\end{aligned}$$

Again, we can use the combinatorial definitions to easily prove some features of multiplication that are very difficult to prove using the recursive definitions. For both of these exercises, you should find a correspondence between the relevant sets.

Exercise 8.7 Show that for all natural numbers m, n, $m \cdot n = n \cdot m$.

Exercise 8.8 Show that for all natural numbers ℓ, m, n, $\ell \cdot (m \cdot n) = (\ell \cdot m) \cdot n$.

8.3 Exponentiation of natural numbers

Exponentiation is an operation that is usually not taught in elementary school, and when it is taught, it tends to be taught only in terms of the recursive definition. However, there is a combinatorial definition as well.

The recursive definition of exponentiation says that $m^0 = 1$ and $m^{S(n)} = m^n \cdot m$. That is, we think of exponentiation as starting with 1, and then multiplying it by m n times. This is probably the way you were taught to understand exponentiation, if at all.

The combinatorial definition again identifies m^n as the unique natural number that has a correspondence to a particular set. This set is the set of all total functions from n to m. Recall, a total function f from n to m is a set of ordered pairs, such that for every $k \in n$, there is exactly one member of f whose first element is k, and such that the second element of any member of f is a member of m. Members of m can appear zero, one, or many times as second elements of ordered pairs in f, but members of n must appear exactly once as first elements.

Exercise 8.9 Show, using the combinatorial definitions, that $m^2 = m \cdot m$. (This involves finding a correspondence between things like $f = \{\langle 0, k\rangle, \langle 1, \ell\rangle\}$ and ordered pairs of elements of m.)

Exercise 8.10 Show, using the combinatorial definition, that the power set of m has 2^m members. (This involves finding a correspondence between functions from m to $\{0, 1\}$ and subsets of m.)

Exercise 8.11 Show, by induction, that the recursive and combinatorial definitions of exponentiation agree.

8.4 Well-orderings

Recall that an ordering R on X is defined as a subset of $X \times X$ that is transitive and irreflexive. That is if $R \subseteq X \times X$ is a relation on X, then (trasitivity) if $\langle x, y\rangle \in R$ and $\langle y, z\rangle \in R$, then $\langle x, z\rangle \in R$, and (irreflexivity) that $\langle x, x\rangle \notin R$.

An ordering R on X is *total* if for every $x, y \in X$, either $x = y$ or $\langle x, y\rangle \in R$ or $\langle y, x\rangle \in R$.

Finally, R is a *well-ordering* if for every non-empty $S \subseteq X$, there is an R-least element of S. That is, there is some $x \in S$ such that for every $y \in S$, if $y \neq x$ then $\langle x, y\rangle \in R$. Once we prove that some relation is a well-ordering, we usually write "$m < n$" rather than "$\langle m, n\rangle \in R$", if it is clear which relation we are using.

Let $R \subseteq \omega \times \omega$ such that $R = \{\langle m, n\rangle \mid m \in n\}$. That is, R is the relation of being a member. We can prove that R is a well-ordering on ω. We have already proved that R is irreflexive (no natural number contains itself). Transitivity and totality are proved by induction.

To prove that it is a well-ordering, we show that if $X \subseteq \omega$ does *not* have an R-least element, then the complement of X contains all the natural numbers. We actually won't do this by proving directly that the complement of X is inductive—instead, we define a new set $X^* = \{n \mid \forall m \leq n(m \notin X)\}$, and we will prove that *this* set is inductive. If X does not have an R-least element, then it is clear that $0 \notin X$ (because it would be R-least in any set that contains it), so $0 \in X^*$ (since every natural number less than or equal to it is not in X). Furthermore, if $n \in X^*$, then neither n nor any smaller natural number is in X. So if $S(n)$ were in X, it would be R-least. So $S(n)$ must not be in X, and thus it must be in X^*. Thus X^* is inductive, and thus contains all natural numbers, so X itself must be empty.

Thus, every non-empty subset of the natural numbers has an R-least element, so this is a well-ordering.

If X and Y are both well-orderings, then we can define a well-ordering on $X \times Y$ by saying that $\langle x_1, y_1\rangle < \langle x_2, y_2\rangle$ if either $y_1 < y_2$ or $y_1 = y_2$ and $x_1 < x_2$. (Note: This is lexicographic ordering where the *second* element in the ordered pair is more significant!) To show that this is a well-ordering, we consider any subset of $X \times Y$ and show how to find the least element of that subset—just look at the set of y that form the second elements of pairs in this set, and find the least such y, and the look at the set of pairs that have this y, and find the least x that forms a pair with it, and the resulting pair must be the least in the set.

Thus, we now have many well-orderings beyond ω, such as $\omega \times \omega$ and $\omega \times (\omega \times \omega)$, each ordered lexicographically.

8.5 Replacement

Axiom (Replacement) If $\varphi(x, y)$ is a formula that defines a function (that is, for every x there is at most one object y satisfying $\varphi(x, y)$), and X is any set, then there is a set $\{y \mid \exists x{\in}X(\varphi(x, y))\}$.

Effectively, this axiom says that for any definable function, we can apply that function to every element of an existing set to get a new set. This axiom wasn't actually part of Zermelo's formulation, but was added by Fraenkel. This axiom expresses some of the idea that as long as a set isn't "too big" it won't lead to paradox—because we just replace the elements of an existing set one for one, the new set can't be "too big" if the old one wasn't.

Using Replacement, Pairing, and Union, we can construct cartesian products without actually using the Power Set axiom. Recall that for any two objects x, y, the Pairing axiom ensures that $\{x\}$, $\{x, y\}$, and $\{\{x\}, \{x, y\}\} = \langle x, y\rangle$ all exist. Note that if we fix a given object y, we can write down a formula stating that one object is the ordered pair consisting of the other object followed by y. This is a formula that defines a function, since there is only one such ordered pair for any first object. We can apply this function to each element $x \in X$ to get a set $X \otimes y = \{\langle x, y\rangle \mid x \in X\}$. Now the relation that fixes X and associates each y to the set $X \otimes y$ is itself another function, and we can use it to replace each element of Y, to get the set $\{X \otimes y \mid y \in Y\}$. Applying the Union axiom to this, we get $X \times Y$.

We can also use Power Set, Empty Set, and Replacement to prove Pairing, without having it as a separate axiom. Using the Empty Set axiom, we know there is a set $\emptyset$. Using Power Set, we know there is a set that contains every subset of this set—and since $\emptyset$ is its own only subset, this means there is a set $\{\emptyset\}$. Using Power Set again, we know there is a set that contains all the subsets of this set, which are just itself and the empty set, so there is a set $\{\emptyset, \{\emptyset\}\}$. Now for any objects z, w, we can write down the formula $\varphi(x, y)$ that states "$(x = \emptyset \& y = z)$ or $(x = \{\emptyset\} \& y = w)$". It is clear that this formula defines a function—for any value of x there is at most one value of y that makes this formula true. We can then apply this function to replace

the elements of $\{\emptyset, \{\emptyset\}\}$, getting $\{z, w\}$. Since z and w were arbitrary, this means that every instance of the Pairing axiom is true.

8.6 Axiomatic ordinals

Well-ordering and Replacement are what we need to define arbitrary functions recursively.

We can define Mostowski collapse. If x is a member of a well-ordering, we define $M(x) = \{M(y) \mid y < x\}$. We can show that this is enough to ensure that $M(x)$ is defined for all members of a well-ordering. If there are any members for which it is *not* defined, we can find the least element of that set. There is a set consisting of all the earlier members, and M is defined for each of those. Thus, we can use the Axiom of Replacement to replace all these earlier members by the output of M, and that gives us a set that is the output of M for the element that was supposed to be the least element for which it wasn't defined.

For addition and multiplication of ordinals, we use the same sets as we used earlier for the natural numbers, and apply Mostowski collapse the lexicographic ordering.

This shows that $2 \cdot \omega \neq \omega \cdot 2$. There is a correspondence, but it is not the same order.

9 The Continuum Hypothesis and further topics

References